UPSC Prelims

Environment & Ecology

Class No. 10

International Conventions Part 1

International Conventions

- **Other climate funds**
- **Adaptation Fund**
- It was established in 2001 to finance concrete adaptation projects and programmes in developing country Parties to the Kyoto Protocol that are particularly vulnerable to the adverse effects of climate change.
- **Green Climate Fund (GCF)**
- It was established in COP 16, in 2010 and developed countries had pledged to mobilise US$ 100 billion per year by 2020 through this fund to support developing countries raise and realize their Nationally Determined Contributions (NDC) ambitions towards low-emissions, climate resilient pathways

International Conventions

- **UN-backed international climate funds**
- **Clean Technology Fund (CTF):**
- It aims at empowering transformation in developing countries by providing resources to scale up low carbon technologies.
- **Climate Investment Funds (CIFs):**
- It aims to accelerate climate action by empowering transformations in clean technology, energy access, climate resilience, and sustainable forests in developing and middle-income countries.
- **UN- Reducing emissions from deforestation and forest degradation (REDD)**: It aims to protect forests, a pre-eminent nature-based solution to the climate emergency.

International Conventions

- **Net Zero Asset Owner Alliance:** *
- It has 29 members, including pension funds, insurance companies, and sovereign wealth funds, and is working on substantial methodologies to align portfolios with net zero Paris targets.
- **International Climate Initiative**
- International Climate Initiative (IKI) is one of the most important instruments of the German Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer protection (BMUV) for the international financing of climate change mitigation and biodiversity.
- IKI operates within the framework of the UNFCCC and the Convention on Biological Diversity (CBD), financing climate change mitigation and biodiversity conservation in developing, emerging and transition countries.

International Conventions

- **Other International Funds**
- **Climate Change Fund of Asian Development Bank (ADB):**

Members outside asia?

- It was established in 2008 to facilitate greater investments in developing member countries (DMCs) to effectively address the causes and consequences of climate change, by strengthening support to low-carbon and climate-resilient development.
- **Forest Carbon Partnership Facility (FCPF):**
- It is a global partnership of governments, businesses, civil society, and Indigenous Peoples focused on reducing emissions from deforestation and forest degradation, forest carbon stock conservation, the sustainable management of forests, and the enhancement of forest carbon stocks in developing countries, activities commonly referred to as REDD+.

International Conventions

- **Other National and local Sources of raising finances**
- **Allocations from National Governments:**
- For example, National Adaptation Fund for Climate Change (NAFCC) is a Central Sector Scheme which was set up in the year 2015-16 to support concrete adaptation activities which mitigate the adverse effects of climate change.
- **Carbon pricing instruments:**
- These include a carbon market approach (where an Emissions Trading Scheme is established, and carbon credits are bought and sold based on a market price per tCO_2e); Carbon emissions tax approach (that can also be in the form of a fossil fuel tax or removal of fossil fuel subsidies) etc.

International Conventions

- **Rio+10**
- As a follow-up to UNCED, the World Summit on Sustainable Development (Rio+10) was held in 2002 in Johannesburg, South Africa.
- Rio+10 affirmed the UN commitment to Agenda 21 alongside the Millennium Development Goals(MDGs). (formed by UNGA)
- MDGs were established following the Millennium Summit of the United Nations in 2000 consisting of 8 goals to be achieved by 2015
- The ensuing Johannesburg Declaration committed the nations of the world towards sustainable development.

International Conventions

- **Rio+20**
- In 2012, the United Nations Conference on Sustainable Development (Rio+20/Rio Earth Summit 2012) was also held in Rio, Brazil.
- The issues touched included:
- checking production of toxic components, such as lead in gasoline and radioactive chemicals,

Green economy

- alternative sources of energy to replace the use of fossil fuels,
- reliance on public transportation, congestion in cities,
- the health problems caused by pollution, and
- the growing usage and limited supply of water.

International Conventions

- The Sustainable Development Goals (SDGs) were born at Rio+20.
- Since 2015, Sustainable Development Goals have been included in the Agenda 2030.
- **Partnership for Action on Green Economy (PAGE)**
- PAGE, launched in 2013, is a direct response to the Rio+20 Declaration, The Future We Want.
- The partnership brings together five United Nations agencies; UNEP, UNDP, ILO, UNIDO (UN Industrial Development Organization), and UNITAR (United Nations Institute for Training and Research)
- It seeks to assist countries in achieving SDG (2030 Agenda), especially SDG 8: "Promote sustained, inclusive and sustainable economic growth, full and productive employment."

International Conventions

- The Dublin Principles pertaining to importance of water as a resource for environmental protection and human development, were presented at the conference.

International Conventions

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Funding?

2 principles

Body?

- **Stockholm+50**
- **Context**
- Recently, Stockholm+50 meeting was held at Stockholm, Sweden to commemorate the 50 years since the 1972 United Nations Conference on the Human Environment (also known as the Stockholm Conference), which made the environment a pressing global issue for the first time.
- Stockholm+50 theme: A healthy planet for the prosperity of all — our responsibility, our opportunity.
- Stockholm+50 agenda:
- To share experiences and initiatives to protect the planet.
- Sustainable recovery from the COVID -19 pandemic.

International Conventions

- **High-level Political Forum on Sustainable Development (HLPF)**
- It is a forum under the UN Economic and Social Council (ECOSOC) tasked with overseeing the outcomes of the 1992 Earth Summit.
- HLPF meets both under the General Assembly every 4 years and the ECOSOC in other years.
- It is responsible for the follow up and the review of progress of implementation of the following:
  - Agenda 21
  - Johannesburg Declaration (Rio+10)
  - Rio+20

UNFCCC

- **United Nations Framework Convention on Climate Change (UNFCCC)** is an international environmental treaty that came into existence under the aegis of the UN.
- It was negotiated in **1992** in New York City and was signed in **1992** at the Rio Earth Summit (UNCED).
- As of Feb 2023, the UNFCCC has **198** parties.
- **Objective:** It stabilises greenhouse gas (GHG) concentrations in the atmosphere at a level that would prevent dangerous consequences.
- **Legal Effect:** It is considered legally non-binding. It provides a framework for negotiating specific international treaties (called "protocols") that aim to set binding limits on greenhouse gases (GHGs). (UNFCCC itself sets no binding limits)

Conferences of the Parties (COP) to UNFCCC

COP 27

NEW

Sharm El Sheikh, Egypt

06 Nov - 18 Nov 2022

View details →

COP 26

Glasgow, Scotland

31 Oct - 12 Nov 2021

View details →

COP 25

Madrid, Spain

02 Dec - 13 Dec 2019

View details →

COP 24

Katowice, Poland

03 Dec - 14 Dec 2018

View details →

COP 23

Bonn, Germany

06 Nov - 17 Nov 2017

View details →

COP 22

Marrakech, Morocco

07 Nov - 18 Nov 2016

View details →

COP 21

Paris, France

30 Nov - 12 Dec 2015

View details →

COP 20

Lima, Peru

01 Dec - 12 Dec 2014

View details →

COP 19

Warsaw, Poland

11 Nov - 23 Nov 2013

View details →

Conferences of the Parties (COP) to UNFCCC

COP 18
Doha, Qatar
26 Nov - 07 Dec 2012
View details →

COP 17
Durban, South Africa
28 Nov - 09 Dec 2011
View details →

COP 16
Cancún, Mexico
28 Nov - 10 Dec 2010
View details →

what else ?

COP 15
Copenhagen, Denmark
07 Dec - 18 Dec 2009
View details →

COP 14
Poznań, Poland
01 Dec - 12 Dec 2008
View details →

COP 13
Bali, Indonesia
03 Dec - 17 Dec 2007
View details →

COP 12
Nairobi, Kenya
06 Nov - 17 Nov 2006
View details →

COP 11
Montreal, Canada
28 Nov - 09 Dec 2005
View details →

COP 10
Buenos Aires, Argentina
06 Dec - 17 Dec 2004
View details →

Conferences of the Parties (COP) to UNFCCC

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COP 09
Milan, Italy
01 Dec - 12 Dec 2003
View details →

COP 08
New Delhi, India
23 Oct - 01 Nov 2002
View details →

COP 07
Marrakech, Morocco
29 Oct - 10 Nov 2001
View details →

COP 06
The Hague, Netherlands
13 Nov - 24 Nov 2000
View details →

COP 05
Bonn, Germany
25 Oct - 05 Nov 1999
View details →

COP 04
Buenos Aires, Argentina
02 Nov - 13 Nov 1998
View details →

COP 03
Kyoto, Japan
01 Dec - 10 Dec 1997
View details →

COP 02
Geneva, Switzerland
08 Jul - 19 Jul 1996
View details →

COP 01
Berlin Mandate
28 Mar - 07 Apr 1995
View details →

Conferences of the Parties (COP) to UNFCCC

- **The COP is the decision-making body of UNFCCC.** All States that are Parties to the Convention are represented at the COP. They review the implementation of any legal instruments that the COP adopts. They promote the effective implementation of the Convention.
- The parties to the convention met annually from **1995 till 2019. The 2020 COP was rescheduled to 2021.**
- Last COP was held in **Sharm-El-Shaikh (Egypt).**

Conferences of the Parties (COP) to UNFCCC

Operating Mechanism of UNFCCC

- **Conference of the Parties (COP):** The supreme decision-making body of the Convention.
- All States that are Parties to Convention are represented at COP.
- It meets every year, unless the Parties decide otherwise.

- **Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol (CMP):**
- Oversees implementation of Kyoto Protocol and takes decisions to promote its effective implementation.
- All States that are Parties to the Kyoto Protocol are represented at the CMP, while States that are not Parties participate as observers.

- **Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA):**
- Oversees the implementation of the Paris Agreement and takes decisions to promote its effective implementation.
- All States that are Parties to the Paris Agreement are represented at the CMA, while States that are not Parties participate as observers.

Kyoto Protocol (COP 3; UNFCCC 1997)

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- The Kyoto Protocol was adopted in Kyoto, Japan, in 1997.
- It came into force in 2005. It is the only global treaty with binding limits on GHG emissions. There are currently 192 Parties to the Kyoto Protocol.
- India ratified in 2002, the USA never ratified, and Canada withdrew in 2012.
- **Goal:** Fight global warming by reducing greenhouse gases (GHGs) concentrations in the atmosphere to "a level that would prevent dangerous anthropogenic interference with the climate system."
- Kyoto protocol is based on the principle of **common but differentiated responsibilities.** It aimed to cut GHG emissions across the developed world by about 5% by 2012 compared with 1990 level.
- The Kyoto Protocol legally binds developed country Parties to emission reduction targets. The Protocol's **first commitment period** started in 2008 and ended in 2012. The **second commitment period** began on 1 January 2013 and will end in 2020.

Kyoto Protocol

- Kyoto Protocol Targets for the first commitment period

- The targets for the first commitment period of the Kyoto Protocol cover emissions of the six main greenhouse gases, namely:
  - **Carbon dioxide (CO_2);**
  - **Methane (CH_4);**
  - **Nitrous oxide (N_2O);**
  - **Hydrofluorocarbons (HFCs);**
  - **Perfluorocarbons (PFCs); and**
  - **Sulphur hexafluoride (SF_6)**

Common but Differentiated Responsibilities

- CBDR (Kyoto Protocol) puts the obligation to reduce current emissions on developed countries on the basis that they are historically responsible for the current levels of GHGs in the atmosphere.

Differentiated Responsibilities

- Under CBDR, developed countries like the US, UK, Russia etc., must contribute more to reduce GHGs by accepting certain binding limits on GHG emissions. They must contribute funds towards reducing GHG emissions to developing and least developed countries (LDCs).
- Developing and least developed countries (LDCs) should do everything they can to cut down their GHG emissions. But nothing is binding on them, and every initiative is voluntary.

Kyoto Protocol

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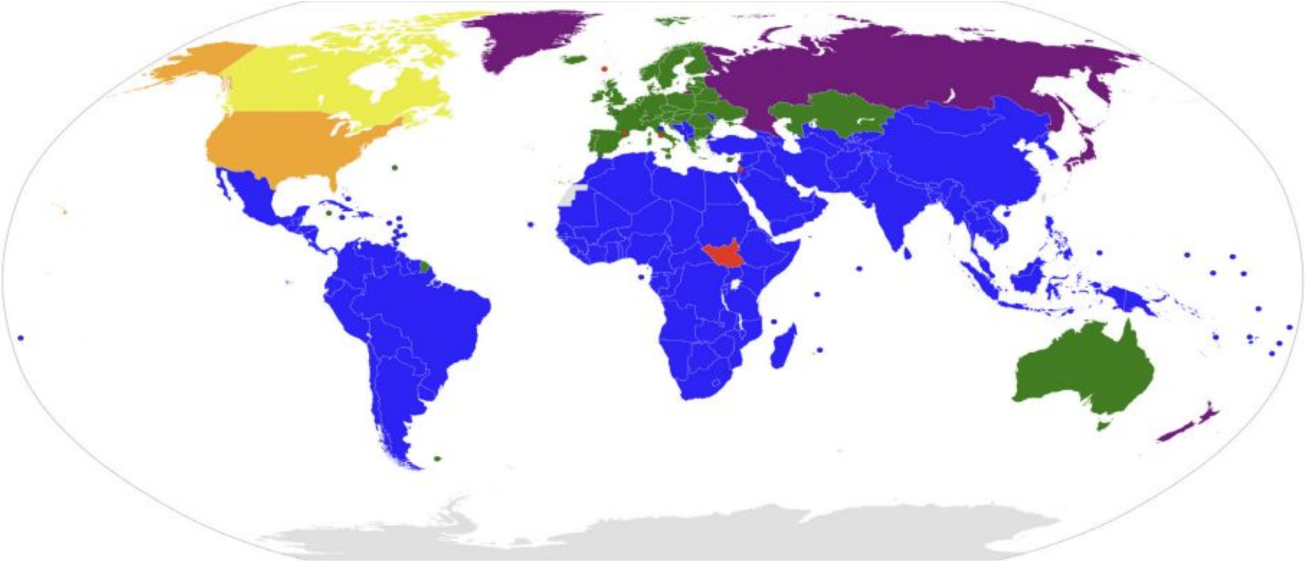
Classification of Parties under the Kyoto Protocol

Annex I	• **Developed countries (US, UK, Russia etc.) + Economies in transition (EIT) [Ukraine, Turkey, some eastern European countries]**
Annex II	• **Developed countries (Annex II is a subset of Annex I)** • Required to provide **financial and technical support** to the **EITs** and **developing countries** to assist them in reducing their greenhouse gas emissions.
Annex B	• **Annex I Parties with first or second-round Kyoto GHG emissions targets (Annex B is a subset of Annex I)** • **Compulsory binding targets** to reduce GHG emissions. • The **first-round Kyoto targets** applied over the years **2008-2012** • The **second-round Kyoto targets** applied from **2013-2020**
Non-Annex I	• Parties to the UNFCCC **not listed in Annex I** (mostly low-income developing countries). • **No binding targets** to reduce emissions.
LDCs	• Least-developed. **No binding targets**.

Kyoto Protocol



- Annex B parties with binding targets in the second period
- Annex B parties with binding targets in the first period but not the second
- Non-Annex B parties without binding targets
- Annex B parties with binding targets in the first period but which withdrew from the Protocol
- Signatories to the Protocol that have not ratified
- Other UN member states and observers that are not party to the Protocol

Flexible Market Mechanisms Under Kyoto Protocol

- Countries bound to Kyoto targets have to meet them largely through domestic action — that is, to reduce their emissions onshore. But they can meet part of their targets through three "market-based mechanisms":

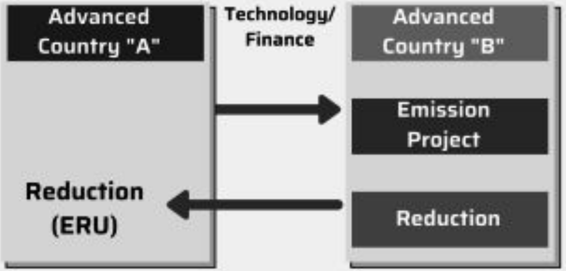
  a. Clean Development Mechanism (CDM)
  b. Emission Trading
  c. Joint Implementation (JI)

Kyoto Protocol

Flexible Market Mechanisms – Kyoto Protocol

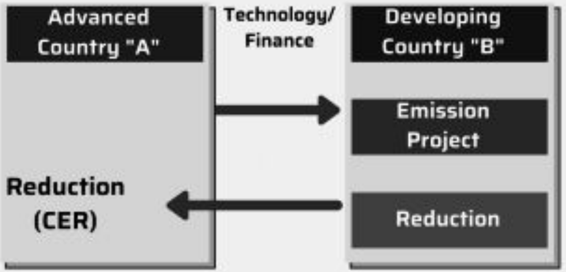
1) Joint Implementation (JI)

Implement emission reduction project between Advanced Countries "A" and Advanced Country "B" and utilize the ERU (Emission Reduction Units) to achieve target for each country.



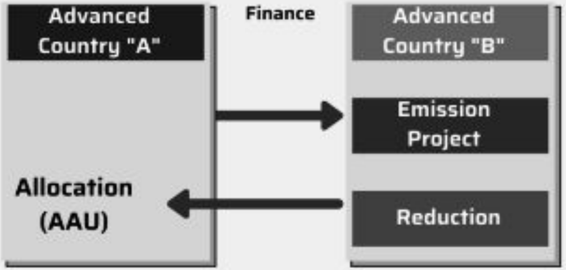
2) Clean Development Mechanism (CDM)

Implement emission reduction project between Advanced Countries "A" and Developing Country "B" and utilize the CER (Certified Emission Reduction) to achieve target for each country.



3) Emission Trading

System to sell and buy AAU (Assigned Amount Unit) among Advanced countries to achieve reduction target.



Kyoto Protocol

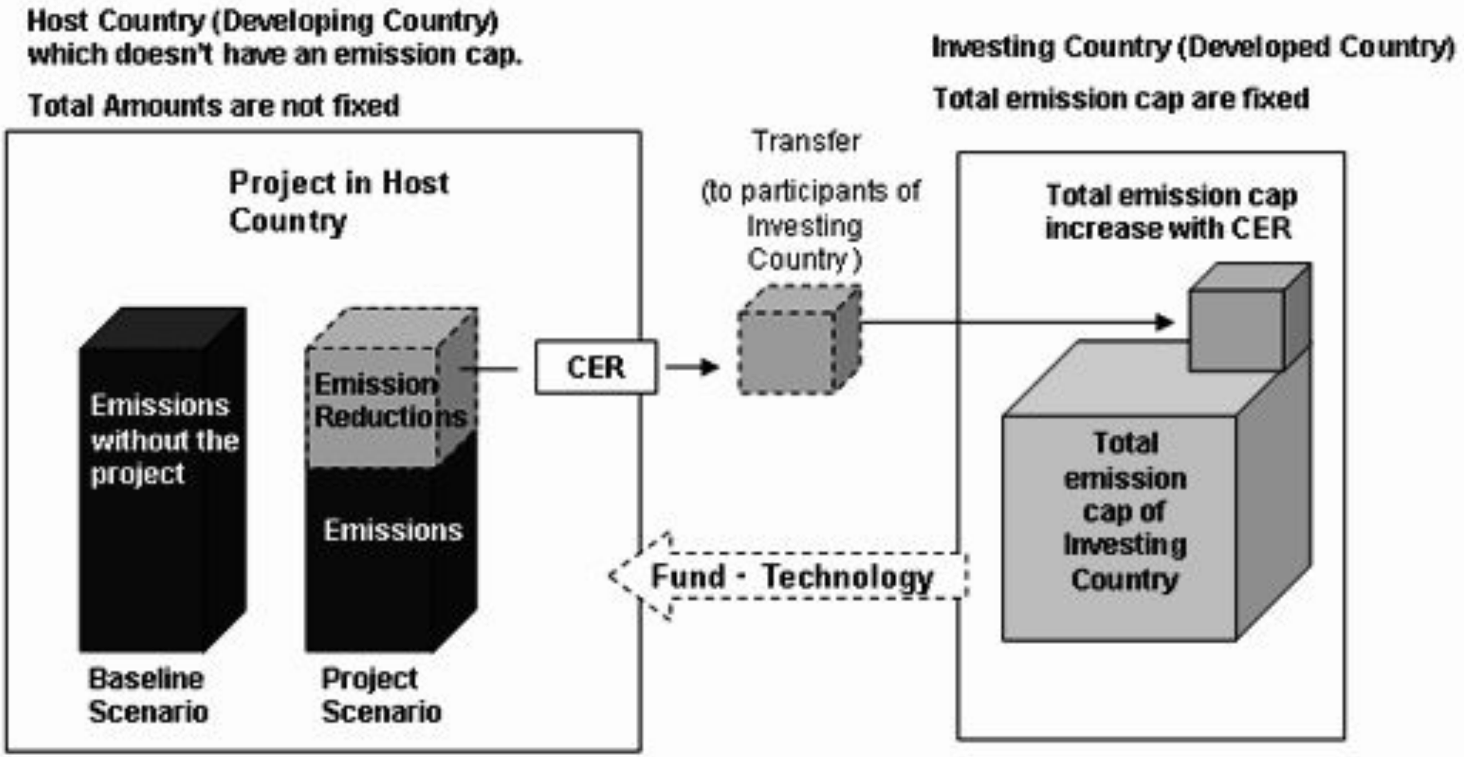
Clean Development Mechanism (CDM)

- CDM allows a country with an emission-reduction commitment under the Kyoto Protocol (Annex B Party like Australia) to implement an emission-reduction project in developing countries (like India).
- Such projects can earn certified emission reduction (CER) credits, each equivalent to one tonne of CO2, which can be counted towards meeting Kyoto targets.
- In simple terms, developed countries emit more and lose carbon credits. They provide financial assistance to developing and LDCs to create clean energy and gain some carbon credits, thereby meeting their Kyoto Quota (Kyoto units) emissions without violations.

Clean Development Mechanism (CDM)

- Suppose a developed country has a Kyoto Quota of 100 Carbon Credits, then it can emit 100 tonnes of CO_2.
- Due to negligence, it emits 110 tonnes of CO_2, i.e., 10 carbon credits are lost (Kyoto Quota violation). The country must compensate for its lost carbon credits to avoid penalties.
- So, It invests some money (equal to 10 carbon credits) in developing and LDCs to build clean energy infrastructure like solar plants, wind farms etc. and will make up for its 10 lost carbon credits and avoid penalty
- Two percent share of proceeds of Certified Emission Reductions (CERs) issued under the Protocol's Clean Development Mechanism projects are used to finance Adaptation Fund.[Adaptation Fund has been discussed in earlier slides]

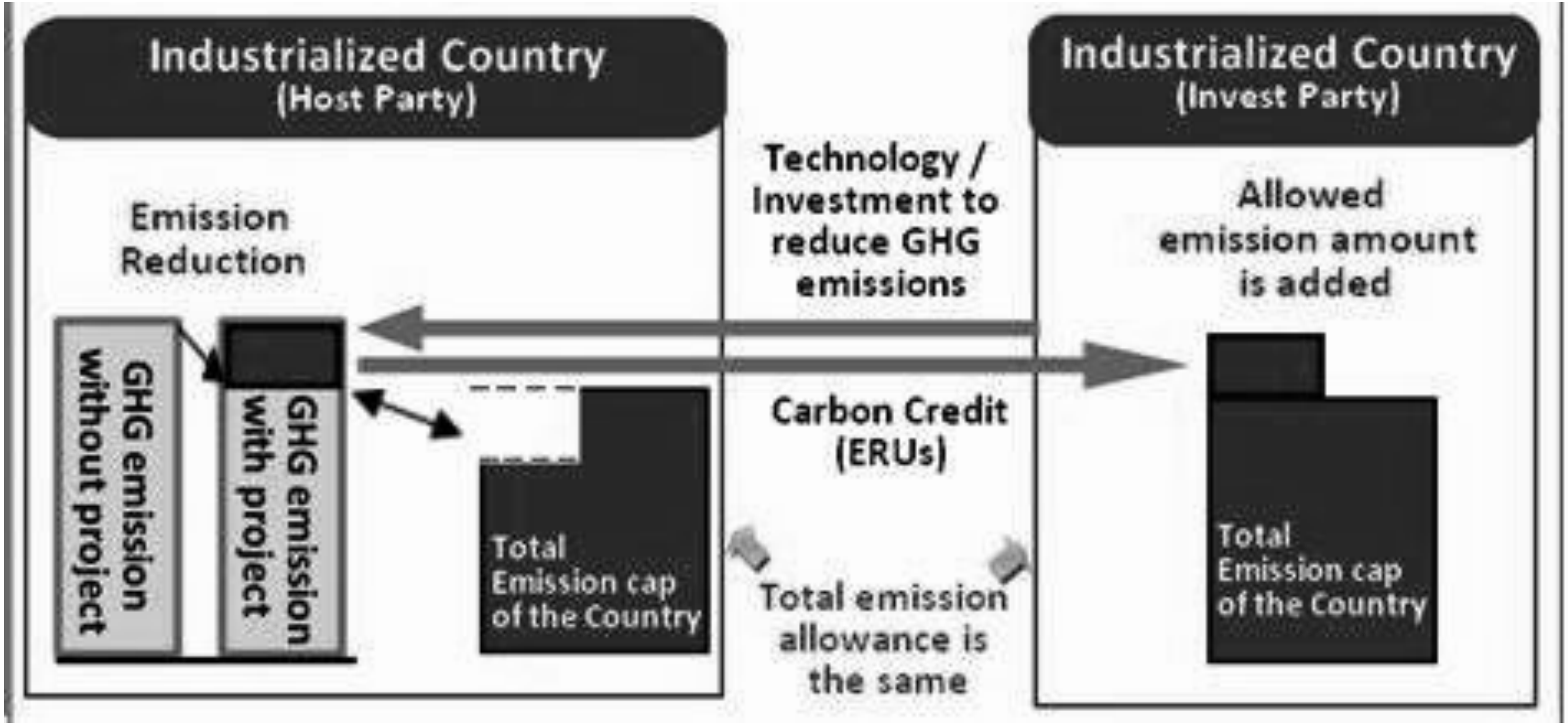
Kyoto Protocol



Joint Implementation (JI)

- The mechanism known as "joint implementation" allows a country with an emission reduction commitment under the Kyoto Protocol (Annex B Party) to earn emission reduction units (ERUs) from an emission reduction project in another Annex B Party, each equivalent to one tonne of CO_2, which can be counted towards meeting its target.
- Joint implementation offers Parties a flexible and cost-efficient means of fulfilling a part of their Kyoto commitments, while the host Party benefits from foreign investment and technology transfer.

Kyoto Protocol



Carbon Credits Trading (Carbon/Emission Trading)

Carbon credit

- A carbon credit (a carbon offset) is a tradable certificate or permit.
- One carbon credit is equal to one tonne of carbon dioxide.
- Carbon credits or offsets can be acquired through afforestation, renewable energy, CO_2 sequestration, methane capture, buying from an exchange (carbon credits trading), etc.

Carbon trading

- Carbon trading is the name given to the exchange of emission permits (carbon credits). This exchange may take place within the economy or may take the form of international transaction.
- Under the Carbon Credits Trading mechanism, countries that emit more carbon than the quota allotted to them buy carbon credits from those that emit less.
- In Carbon trading, one credit gives the country or a company the right to emit one tonne of CO_2.
- Carbon credits are traded at various exchanges across the world. The Multi-Commodity Exchange of India launched futures trading in carbon credits in 2009.

Kyoto Protocol

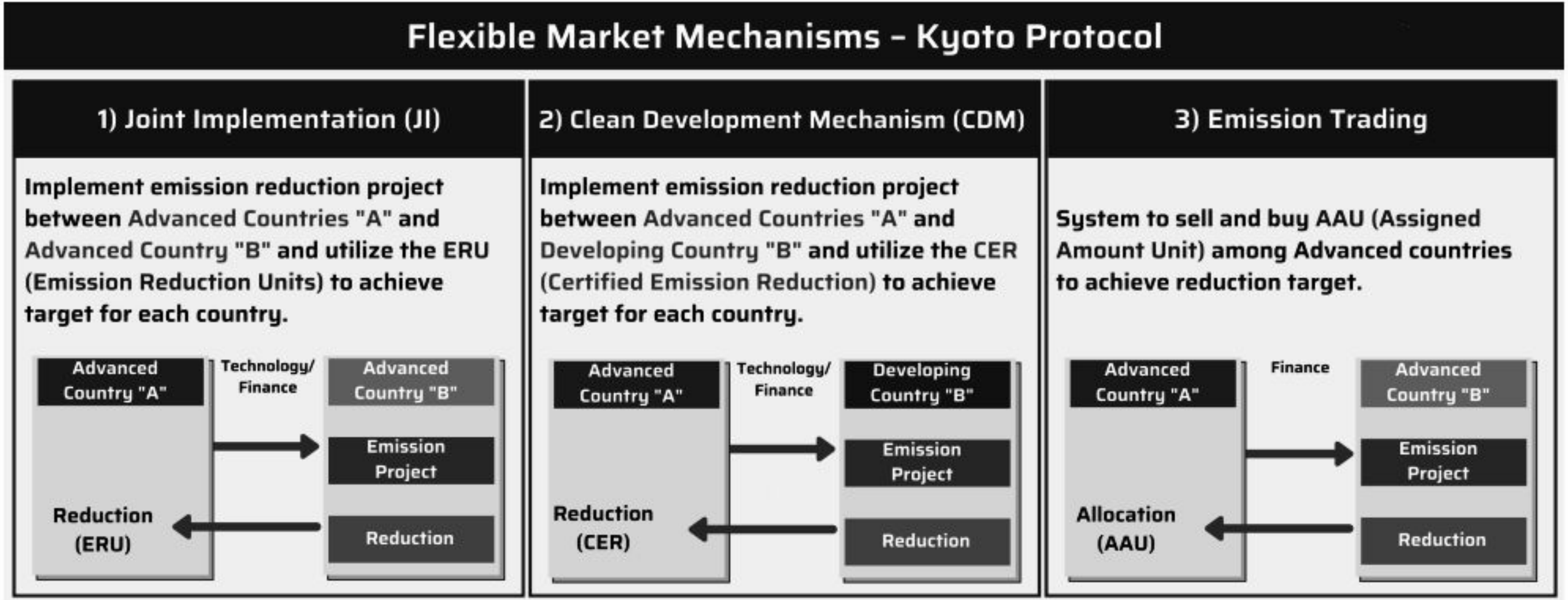
What is traded under Emission Trading Mechanism? → JI / CDM

- Parties with commitments under the Kyoto Protocol (Annex B Parties) have accepted targets for limiting or reducing emissions. These targets are expressed as levels of allowed emissions, or assigned amounts,at over the 2008-2012 commitment period. The allowed emissions are divided into assigned amount units (AAUs).
- Emissions trading, as set out in Article 17 of the Kyoto Protocol, allows countries that have emission units to spare - emissions permitted them but not "used" - to sell this excess capacity to countries that are over their targets.

Other Traded Units:

- A **removal unit (RMU)** on the basis of land use, land-use change and forestry (LULUCF) activities such as reforestation
- **Emission reduction unit (ERU)** generated by a joint implementation project
- **Certified emission reduction (CER)** generated from a clean development mechanism project activity

Kyoto Protocol



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Types of Carbon Trading

Emission trading/'cap-and-trade'	Offset Trading/Carbon Project/'baseline-and credit'
Sell/purchase	Create
Emissions trading allows countries to sell unused emission units to countries that have exceeded their targets. Carbon is tracked and traded like any other commodity in a "carbon market."	It is to be earned by a country by investing some amount of money in such projects, known as carbon projects, which will emit a lesser amount of GHGs into the atmosphere

Types of Carbon Trading

Emission trading/'cap-and-trade'	Offset Trading/Carbon Project/'baseline-and credit'
Carbon offsetting is effective if projects are additional, permanent, meet certain key criteria and project standards, and do not engage in greenwashing.	Improper reporting and discrepancies in maximum GHG levels between countries limits cap and trade effectiveness on a global scale.
Trading of Assigned Amount Units amount to this.	Trading of Removal unit (RMU), Certified Emissions Reductions (CER) will be included here.

Kyoto Protocol

Non-Compliance of Kyoto and Penalties

- If a country does not meet the requirements for measurements and reporting, the country loses the privilege of gaining credit through joint implementation projects.
- If a country goes above its emissions cap and does not try to make up the difference through any of the mechanisms available, then said the country must make up the **difference plus an additional thirty per cent during the next period.**
- The country could also be **banned** from participating in the **'cap and trade' program**

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Bali (Indonesia) CCC 2007 (COP 13; CMP 3)

- Governments adopted the **Bali Road Map**
- The Bali Road Map included the **Bali Action Plan**, which charts the course for a new negotiating process designed to tackle climate change.
- The Bali Action Plan was a comprehensive process to enable the full, effective and sustained implementation of the Convention through long-term cooperative action, now, up to and beyond 2012.

Poznan (Poland) CCC 2008 (COP 14; CMP 4)

- It launched the Adaptation Fund under the Kyoto Protocol.
- The Fund is financed partly by the government and private donors and also from a 2% share of proceeds of Certified Emission Reductions (CERs) issued under Clean Development Mechanism projects.

Copenhagen (Denmark) CCC 2009 (COP 15; CMP 5)

- The Copenhagen Accord included the goal of limiting the maximum global average temperature increase to **no more than 2°C** above pre-industrial levels, subject to review in 2015.
- Developed countries promised to provide **US$30 billion** for the period 2010-2012 and to mobilize long-term finance of a further US$100 billion a year by 2020.

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Cancún (Mexico) CCC 2010 (COP 16; CMP 6)

- Parties agreed to commit to a maximum temperature rise of 2°C above pre-industrial levels and to consider lowering that maximum to 1.5°C in the near future.
- Parties agreed to establish a **Green Climate Fund (GCF)** to finance projects, programmes, policies and other activities in developing countries via thematic funding windows.
- Government of India also agreed to include **carbon capture and storage (CCS) in the projects under the Clean Development Mechanism (CDM),** subject to technical and safety standards

Durban (Denmark) CCC 2011 (COP 17; CMP 7)

- Decided to adopt a universal legal agreement on climate change no later than 2015.
- **Second phase of the Kyoto Protocol was secured.**
- Approved the Governing Instrument for the Green Climate Fund (GCF)

Post Kyoto Protocol UNFCCC Summits

Doha (Qatar) CCC 2012 (COP 18; CMP 8) → Doha amendment.

- COP18 reached an agreement to extend the life of the Kyoto Protocol, which had been due to expire at the end of 2012 until 2020 (second commitment period 2013-20).
- The extension of the Kyoto Protocol until 2020 limited in scope to **only 15% of the global CO2 emissions.** This was due to the lack of participation of Canada, Japan, Russia, Belarus, Ukraine, and New Zealand (they all refused to join the second commitment period under the Kyoto Protocol) and the United States (never ratified Kyoto).
- Also, developing countries like China, India and Brazil are not subject to emission reductions under the Kyoto Protocol.

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Warsaw (Poland) CCC 2013 (COP 19; CMP 9)

- The term **Intended Nationally Determined Contributions (INDC)** was coined in Warsaw.
- Further, the **Warsaw Mechanism was proposed**, which would provide expertise, and possibly aid, to developing nations to cope with **loss and damage** from such natural extremities as heatwaves, droughts and floods and threats such as rising sea levels and desertification.

Lima (Peru) CCC 2014 (COP 20; CMP 10)

- The overarching goal of the conference is to reduce GHGs **to limit the global temperature increase by 2030 to 2°C above the 1850 baseline** or pre-industrial era.
- The agreement urged parties to take national pledges by finalising their **Intended Nationally Determined Contributions (INDC)** by November 2015 (before Paris Summit).

Paris Agreement

Why Paris Deal?

- All the significant agreements and decisions were reserved for Paris Summit 2015. It is the most crucial conference post-Kyoto because of the INDC commitments made by major polluters.
- The conference's objective is to achieve a legally binding and universal agreement on climate to be signed in 2015 and implemented by 2020.
- Before the conference, 146 national climate panels publicly presented draft national climate contributions (INDCs).
- However, no detailed timetable or country-specific goals for emissions were incorporated into the Paris Agreement (as opposed to the previous Kyoto Protocol with specific deadlines).

Paris Agreement

- COP21 negotiated the Paris Agreement, a global agreement on reducing climate change. It entered into force in November 2016 after ratification by 55 countries that account for at least 55% of global emissions had been met.
- **Signatories:** 195 as of 2023; 180+ countries have ratified; India signed and ratified in 2016.
- The expected key result was an agreement to set a goal of limiting global warming to "well below 2°C" compared to pre-industrial levels.
- The agreement calls for zero net anthropogenic GHG emissions to be reached during the second half of the 21st century.

Paris Agreement

- According to the adopted version of the Paris Agreement, the parties will also "**pursue efforts to limit the temperature increase to 1.5 °C.**" The 1.5 °C goal will require zero-emissions sometime between 2030 and 2050.
- The developed countries reaffirmed the commitment to mobilise **$100 billion** a year in climate finance by 2020 and agreed to continue it further until 2025.
- In 2017, the US announced that it would cease all participation in the 2015 Paris Agreement. In accordance with Article 28 of the Paris Agreement, it remained a signatory till November 2020.

↳ US also pulled out of?

Paris Agreement

Paris Rulebook

ChatGPT

- The Paris rulebook is a legally binding international treaty on climate change.

- The Paris Rulebook is a set of guidelines and procedures that were adopted at the 24th Conference of Parties (COP 24) to the United Nations Framework Convention on Climate Change (UNFCCC) in 2018. The Paris Rulebook provides details on how countries will measure, report, and verify their greenhouse gas emissions reductions and their progress towards meeting the goals set out in the Paris Agreement.

- The Paris rule book governs how the world community of 191 countries must pledge emissions reduction targets under the Paris Agreement and report on their progress. In signing the Paris rulebook, the 191 countries agreed that they would limit the rise in global temperatures to 1.5-2°C by 2030.

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Global Stocktake

- The Global Stocktake is a process established under the Paris Agreement on climate change. It is a periodic review of collective progress towards achieving the agreement's long-term goals, including the goal of keeping global temperature rise well below 2 degrees Celsius above pre-industrial levels and pursuing efforts to limit it to 1.5 degrees Celsius.
- The Global Stocktake is designed to take place **every five years**, starting in 2023, and will assess the collective progress of all countries in implementing their nationally determined contributions (NDCs) and the overall progress towards achieving the goals of the Paris Agreement.

Article 6 of the Paris Agreement and Carbon Pricing

- Under Article 6 of the Paris Agreement, countries agreed to set up a new global carbon market system (for trading of carbon emissions) to help countries decarbonise their economies at lower cost.
- Article 6 aims at promoting integrated, holistic and balanced approaches that will assist governments in implementing their NDCs through voluntary International cooperation.

Article 6 of the Paris Agreement and Carbon Pricing

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Article 6 contains three separate mechanisms for **"voluntary cooperation":**

- **International Transfer Mitigation Outcomes (ITMO):** Article 6.2 (Market based mechanism) allows countries to strike direct bilateral and voluntary agreements to trade carbon units (that are in excess of their NDCs Targets).
- **Sustainable Development Mechanism (SDM):** Article 6.4 (Market based mechanism) creates a centralized governance system for public and the private sector to trade emissions reduction anywhere in the world. , is due to replace the Clean Development Mechanism (CDM).
- **Non Market based mechanism:** Article 6.8 develops a framework for cooperation between countries to reduce emissions outside market mechanisms, such as developmental aid (ie. where no trade is involved). This cooperation mechanism, if properly designed, should make it easier to achieve reduction targets and raise ambition.
- These systems are more transparent than mechanisms under Kyoto Protocol.

Article 6 of the Paris Agreement and Carbon Pricing

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- In particular, Article 6 could also establish a policy foundation for an emissions trading system, which could help lead to a global price on carbon.

- Under this mechanism, countries with low emissions would be allowed to sell their exceeding allowance to larger emitters, with an overall cap of greenhouse gas (GHG) emissions, ensuring their net reduction.

- Supply and demand for emissions allowances would lead to the establishment of a **global carbon price** that would tie the negative externalities of GHG emissions to polluters. In other words, by paying a price on carbon, states exceeding their NDCs would bear the costs of global warming.

tax on -ve externalities.

Polluter pays

Article 6 of the Paris Agreement and Carbon Pricing

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- **Resolution of issue of double counting (at CoP 26)**
- Double counting refers to the situation where two countries, or more generally speaking, two parties, claim the same carbon removal or emission reduction.
- The issue of double counting from now on will be addressed by the means of **"corresponding adjustments"** of national carbon inventories.
- This means that if one country uses a carbon offset to reduce its carbon footprint, the credit cannot be claimed by another nation.
- The Article 6 rules thereby allow the nation, which is hosting the offsetting project, to make the ultimate decision whether the emission reductions will be counted towards its own targets or sold to another nation.
- In any way, the country hosting the offsetting project has to notify a UN supervisory board.

Carbon Tax

- A carbon tax imposes a tax on each unit of GHG emissions and gives countries an incentive to reduce pollution whenever doing so would cost less than paying the tax.

- A carbon tax is proposed in many developed and developing countries. However, The proposal has faced severe political resistance (politician – corporate nexus; people fear more burden).

Carbon Tax

- Budget of 2010-11 introduced a Clean Energy Cess of Rs. 50 per tonne on both domestically produced and imported coal. Later it was increased to Rs. 100.
- With the introduction of the Goods and Service Tax (GST), the Clean Energy Cess was abolished.
- A new cess on coal production, called the GST Compensation Cess of Rs. 400 per tonne is put in place. This cess is used to raise revenues for the National Clean Energy Fund.

National Clean Energy Fund (NCEF)

- NCEF is a fund created in 2010-11 to allocate Clean Energy Cess (later GST Compensation Cess) for funding research and innovative projects in clean energy technologies of public or private sector entities, up to the extent of 40% of the total project cost.
- The Fund is designed as a non-lapsable fund under Public Accounts and with its secretariat in the Department of Expenditure, Ministry of Finance.

Carbon Tax

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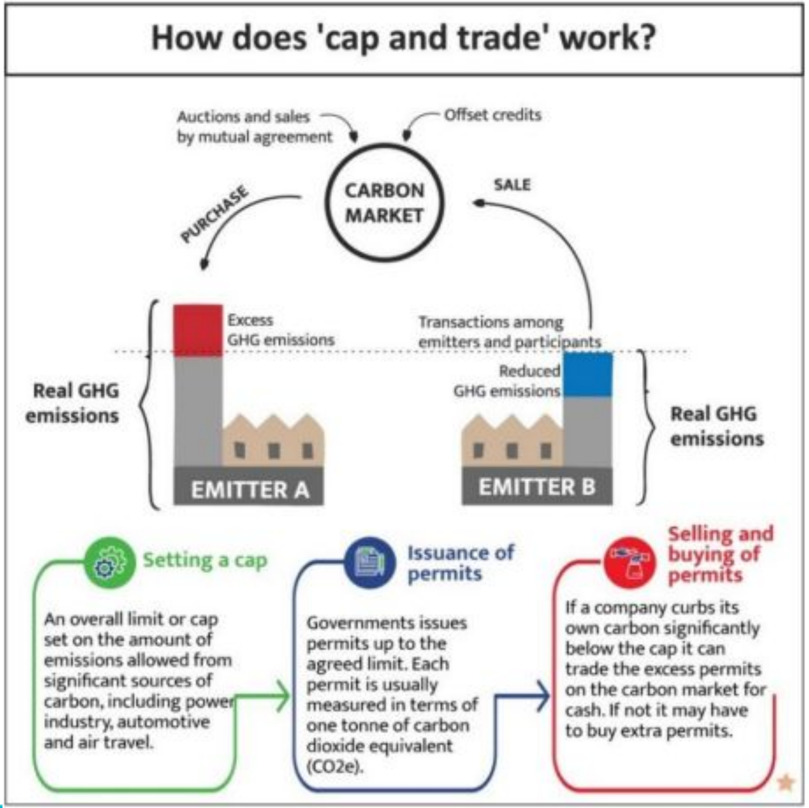
Carbon Tax (price instrument)	Carbon Trading (quantity instrument)
A pure carbon tax fixes the price of carbon but allows the amount of carbon emissions to vary.	A pure carbon cap places a limit on carbon emissions, letting the market price of carbon credits vary.
A carbon tax directly establishes a price on GHGs. (One has to pay for every ton of emissions)	Carbon Trading program issues a set number of emissions "allowances" each year. (One has to pay only after they exceed their emissions quota)

Carbon Tax

Carbon Tax (price instrument)	Carbon Trading (quantity instrument)
Carbon tax ensures everyone knows the price being paid for each unit of carbon dioxide emitted, but uncertainty remains about the actual quantity of emissions.	Conversely, cap-and-trade provides certainty about the quantity of emissions (it cannot exceed the cap), but uncertainty about the cost of achieving these reductions.
A carbon tax offers stable carbon prices, so energy producers and entrepreneurs can make investment decisions without fear of fluctuating regulatory costs	The regulatory costs are always fluctuating, and hence making investment decisions is fraught with risks.

Carbon Trading



Carbon Trading

Existing carbon trading mechanisms in India

Renewable Energy Certificate (REC)

- A market-based instrument to promote renewable energy and facilitate compliance of renewable purchase obligations (RPO) by obligated entities.
- **One REC is treated as equivalent to 1 MWh.**
- **Can also be bought voluntarily by entities.**
- 2 categories of RECs: **solar RECs and non-solar RECs.**
- **Traded only on the Central Electricity Regulatory Commission (CERC) approved power exchanges** like- Indian Energy Exchange (IEX) and Power Exchange India (PXIL).

Energy Saving Certificates (ESCerts) under Perform, Trade and Achieve (PAT) scheme

- Market based regulatory instruments to reduce the specific energy consumption (SEC) in energy intensive industries.
- **EScerts are issued by Central Government** for the difference of quantity between notified target and achieved SEC.
- Covers **13 sectors (Designated Consumers)** including- Aluminium, DISCOMs, Hotels (under commercial buildings), Cement, Fertilizer, Pulp & Paper, Thermal Power Plant, Iron & Steel etc.
- Traded on **Indian Energy Exchange (IEX) and Power Exchange India Limited (PXIL).**

Carbon Trading

Digital Monitoring, Reporting and Verification (D-MRV) Systems

- With growing interest of countries to participate in carbon markets to meet their climate change goals, innovative D-MRV are evolving to track reductions in GHG emissions.
- Represent the first step in end-to-end digitalization of post-2020 carbon markets.
- Based on AI, machine learning, blockchain, smart sensors, drones etc.

Carbon Trading

- The central government is considering implementation of a carbon trading scheme that will subsume all carbon trading mechanisms in India.

- The government proposes to begin with a voluntary carbon market and gradually shift to 'cap and trade', where industries are given emission targets like in EU emission trading system markets.

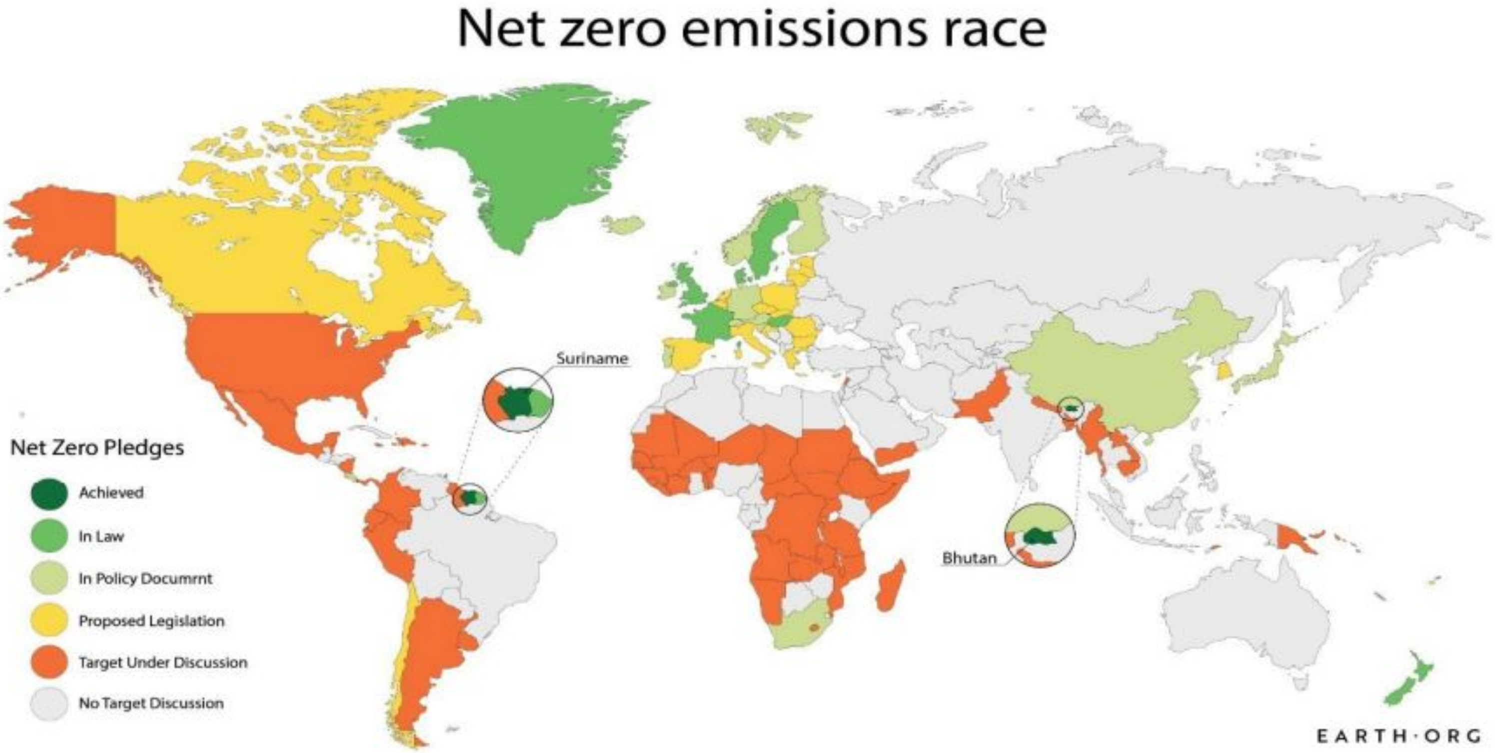
Carbon Neutrality

- **Net zero** means the amount of GHGs emitted into the atmosphere is less than the amount taken out.
- Similarly, **Carbon Neutrality** (Net Zero Carbon Footprint) refers to achieving net zero CO2 emissions.
- The global emissions of GHGs need to be cut by 45 per cent by 2030 and become net zero by 2050 to keep warming below 1.5°C as decided under the Paris Agreement.

How Can Net Zero Emissions Be Achieved?

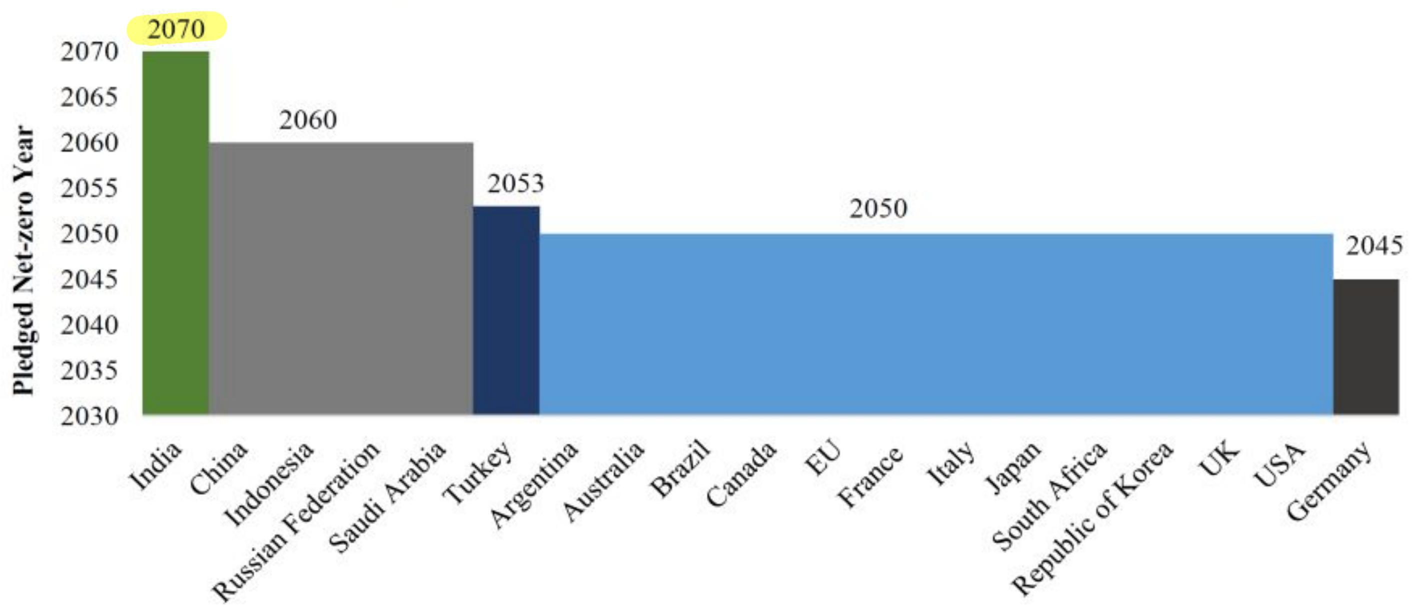
- Clean coal technology to reduce CO2 in the atmosphere (short-term measure). Carbon sequestration (long-term sustainable measure)
- Afforestation: Forests as carbon Sinks (long-term sustainable measure)
- Geoengineering to fight climate change (last resort superhuman options)
- Electric Mobility and Renewable Energy (prevention is better than cure option)
- Carbon Trading and Carbon Tax (long-term measure)
- To contain carbon emissions, UNFCCC summits have arrived at the "polluter pays" principle by placing a price on carbon dioxide and other greenhouse gas emissions.
- The "polluter pays" principle can be implemented through a Carbon Tax or Carbon Trading.

Carbon Neutrality



Carbon Neutrality

Carbon Neutrality



Carbon Neutrality

- The nondescript hamlet of Palli in Jammu and Kashmir's border district of Samba became country's first **"carbon neutral panchayat"**.

- 500 kV solar plant will provide clean electricity to houses in the model panchayat.

- The village, which has 450 houses, have been given solar 'Chulhas' (stoves) and all the houses.

Carbon Neutrality

- **Climate Neutral Now**
- Climate Neutral Now was launched by the **UNFCCC secretariat** in **2015.** It is aiming at encouraging and supporting all levels of society to take climate action to achieve a climate-neutral world by mid-century, as enshrined in the Paris Agreement.
- Climate neutrality is a three-step process, which requires individuals, companies and governments to:

1. Measure their climate footprint;
2. Reduce their emissions as much as possible;
3. Offset what they cannot reduce with UN-certified emission reductions

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Carbon Border Adjustment Mechanism (CBAM)

- It is a plan from the **European Union (EU)**, proposed in 2021.
- It will **tax carbon-intensive products,** such as iron and steel, cement, fertilizer, aluminium, electricity, and hydrogen, from **2026.**
- It aims to eliminate the difference in carbon price paid by companies subjected to the EU's Emissions Trading System (ETS) or its domestic compliance-based carbon market.
- It is the part of the "Fit for 55 in 2030 package".
  - **Fit for 55:** EU's plan to reduce GHG emissions by at least 55% by 2030 compared to 1990 levels, in line with the European Climate Law.

Carbon Border Adjustment Mechanism (CBAM)

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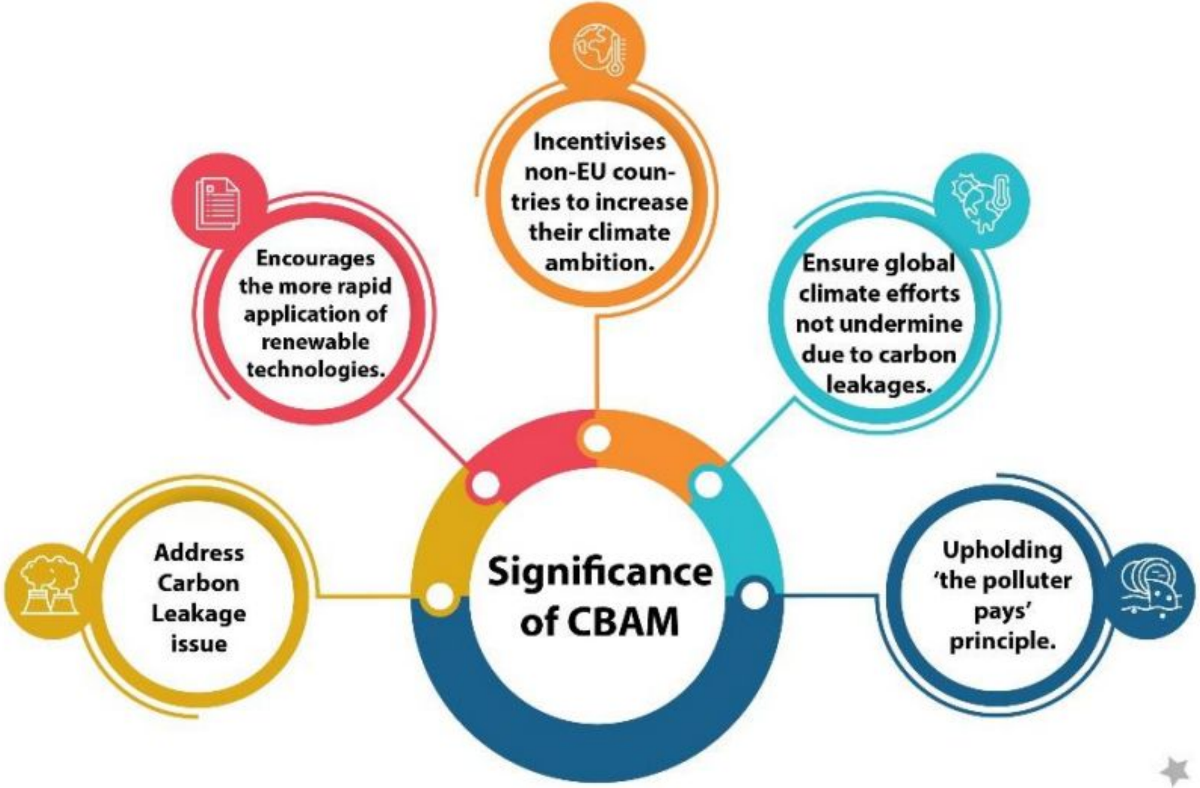
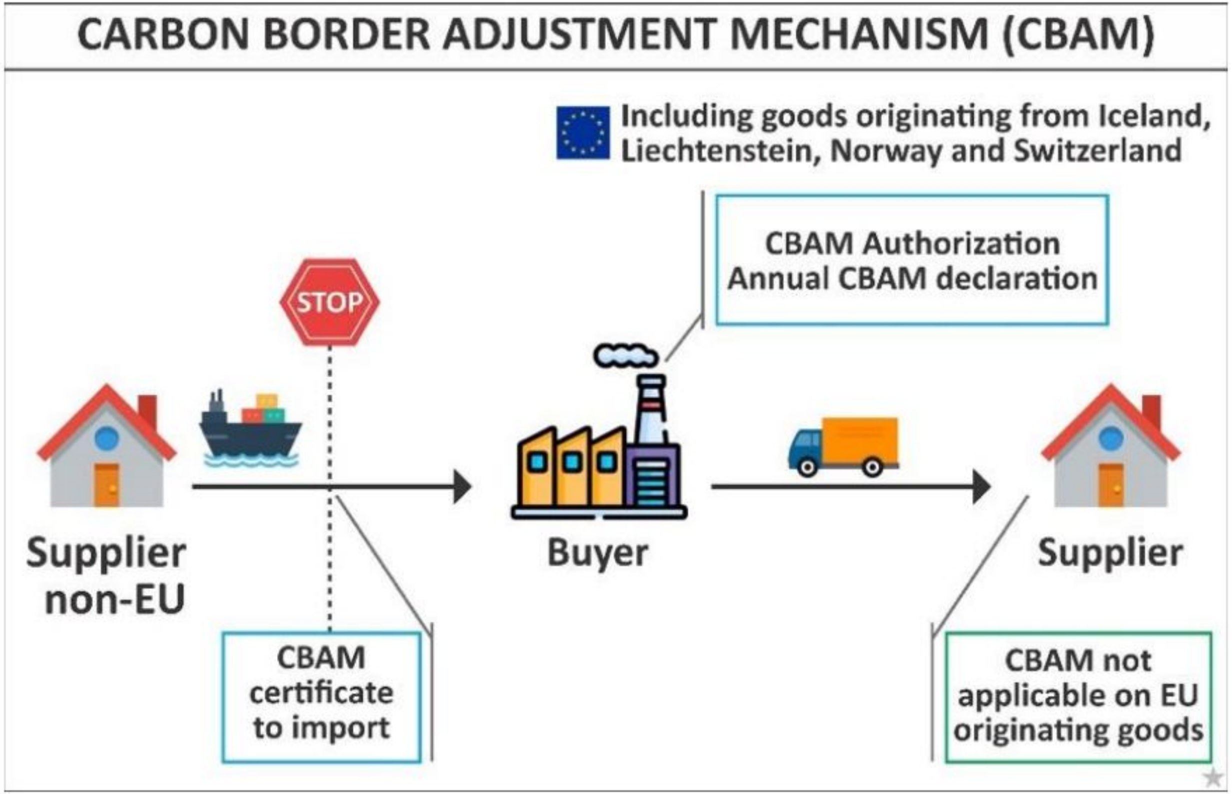
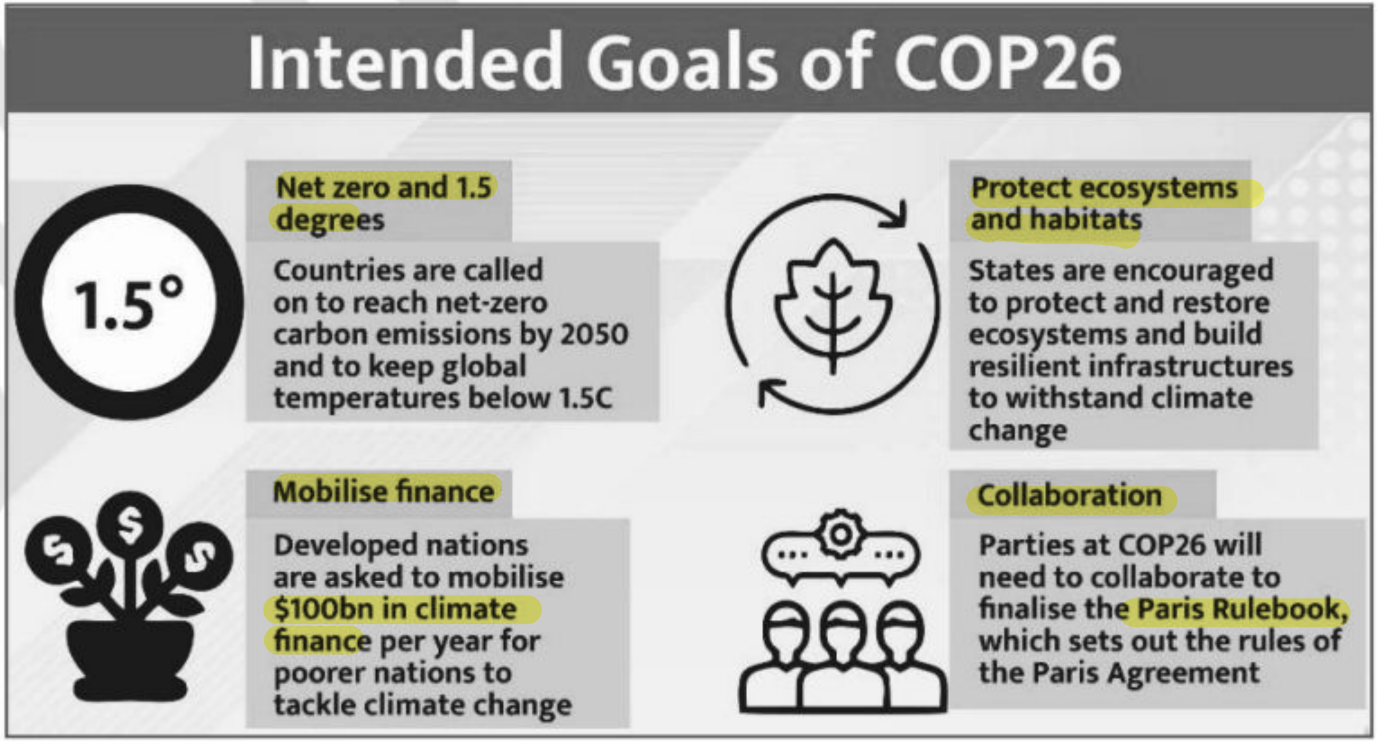


Carbon Border Adjustment Mechanism (CBAM)



India
↓
opposed

COP26 of UNFCCC



Initiatives signed by India at COP26

Name	Details
Breakthrough Agenda	• It commits countries to work together to **make clean technologies and sustainable solutions the most affordable, accessible and attractive option in each emitting sector** (power, road transport, steel, agriculture etc.) globally before 2030.
Declaration on "accelerating the transition to 100% zero-emission cars and vans"	• Non-binding agreement among governments, automotive manufacturers, financial institutions, and civil society organisations. • **Aim:** To work towards **all sales of new cars and vans being zero emission globally by 2040,** and by no later than 2035 in leading markets.

Major Initiatives launched at COP26

GLOBAL RESILIENCE INDEX INITIATIVE (GRII)

- Launched by 10 global organisations. (Including Coalition for Disaster Resilience Infrastructure, the United Nations Office for Disaster Risk Reduction.)

Goals of GRI:

- Offer global open reference risk data using metrics built on insurance risk modelling principles;
- Provide shared standards and facilities applicable to a wide range of uses,
- including corporate climate risk disclosure, national adaptation planning and reporting, and the planning of prearranged humanitarian finance.

Major Initiatives launched at COP26

INFRASTRUCTURE FOR RESILIENT ISLAND STATES (IRIS)

- Launched by India, along with Australia, UK, Fiji, Jamaica and Jamaica.
- It has been co-created by the Coalition for Disaster Resilient Infrastructure (CDRI) with support from Member Countries and organizations and Small Island Developing States (SIDS) representatives.
- It is a dedicated initiative that will provide technical support on the multifaceted issues posed by infrastructure systems and promote disaster and climate resilience of infrastructure assets in SIDS, and share latest knowledge and learnings targeted to specific infrastructure sectors.

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INFRASTRUCTURE FOR RESILIENT ISLAND STATES (IRIS)

IRIS targets to deliver three intended outcomes as:

- Improve resilience of SIDS infrastructure to climate change and disaster risk.
- Strengthen knowledge and partnerships for integrating resilience in SIDS infrastructure.
- Promote gender equality and disability inclusion through resilient SIDS infrastructure.

Small island developing states (SIDS), comprising countries in the Caribbean, Pacific, Atlantic, Indian Ocean, Mediterranean and South China Sea regions, are among the most vulnerable to geophysical and hydro-meteorological hazards.

Major Initiatives launched at COP26

THE GREEN GRIDS INITIATIVE-ONE SUN ONE WORLD ONE GRID PROJECT

- It was a multi-layered dialogue between the OSOWOG initiative by India in partnership with COP26 Presidency, International Solar Alliance (ISA), World Bank, UK government and Wilton Park.
- It is India's initiative to build a global ecosystem of interconnected renewable energy resources that connects 140 countries through a common grid.
- The blueprint for the OSOWOG has been developed under the World Bank's technical assistance programme.
- **Implementation is divided into three main phases**
  - **Phase 1** ensures interconnectivity in the Asian continent.
  - **Phase 2** connects the functional first phase to the pool of renewable resources in Africa.
  - **Phase 3** aims to achieve a global interconnection.

COP27 of UNFCCC

- It aims to build on previous successes, including the Glasgow Climate Pact of COP26 and pave the way for higher ambition on mitigation, adaptation and climate finance, with focus on loss and damage.
- It also included the 17th session of the COP serving as the meeting of the Parties to the Kyoto Protocol (CMP 17), and the fourth session of the COP serving as the meeting of the Parties to the Paris Agreement (CMA 4).
- Concluded with release of Sharm el-Sheikh Implementation Plan.

COP27 of UNFCCC

Areas of discussion	Important Decisions and Developments
Climate targets	• Countries requested to **revisit and strengthen their 2030 climate targets by the end of 2023,** as necessary to align with the Paris Agreement.
Mitigation	• **Finalised the details of Mitigation work programme** to urgently scale up mitigation ambition and implementation in this decade.
Adaptation	• **Development of a framework for the global goal on adaptation** to be undertaken through a structured approach under the **Glasgow–Sharm el-Sheikh work programme in 2023 at COP28.** • **New pledges, totalling more than USD 230 million,** made to the Adaptation Fund.
Finance	• **Sharm el-Sheikh dialogue launched on Article 2.1c** of the Paris Agreement to report back at COP28. ○ **Article 2.1c** – reads "financial flows" should be aligned with global temperature targets.

COP27 of UNFCCC

Areas of discussion	Important Decisions and Developments

Loss and damage	• **New funding arrangements established for loss and damage.**
	  ○ Will assist developing countries that are particularly vulnerable to the adverse effects of climate change.
	  ○ **A transitional committee** to be set up to make recommendations for the operationalization of the new funding arrangements at COP28.
	• **Institutional arrangements for operationalization of the Santiago network.**
	  ○ The host of the secretariat of the network will be selected by 2023.

Energy	• Call to transition towards **low-emission energy systems.** • Call to accelerate efforts towards the **phasedown of unabated coal power** and **phase-out of inefficient fossil fuel subsidies.**
Financial system reform	• **Multilateral development banks and international financial institutions encouraged to reform their operational model, channels and instruments** to address the global climate emergency.

COP27 of UNFCCC

Areas of discussion	Important Decisions and Developments
Just transition	• **A work programme on just transition launched.** • Includes annual "high-level ministerial round tables", with the first taking place at COP28 next year.
Agriculture	• **Koronivia Joint Work for Agriculture (KJWA) given another four-year lease** by establishment of the four-year Sharm el-Sheikh joint work on implementation of climate action on agriculture and food security. o KJWA is the **only programme to focus on agriculture and food security under UNFCCC.**
Technology transfer and deployment	• **First joint work programme of the Technology Executive Committee** and the **Climate Technology Centre and Network**, set up for 2023–2027.
Other	• First COP cover decision to mention **food, rivers, nature-based solutions, tipping points and the right to a healthy environment.**

Initiatives launched at COP27 of UNFCCC (Read Once)

Initiative	Launched by	Objective and other information
Agriculture and food		
Agricultural Innovation Mission for Climate (AIM4C or AIM4Climate)	**UAE & US**  Not a Member	• Aim: To **boost investment and other support for climate-smart agriculture and food systems** innovation over five years (2021-25) to the tune of $8bn.
Initiative on Nutrition and Climate Change (I-CAN)	**Egypt as COP27 Presidency, in partnership with WHO, FAO,** other UN agencies and partners.	• Aim: To support Member States in delivering climate change adaptation and mitigation policy action which simultaneously- o **improves nutrition** and o **triggers transformative action to deliver healthy diets** from sustainable food systems

Initiatives launched at COP27 of UNFCCC (Read Once)

Initiative	Launched by	Objective and other information
Agriculture and food		
International Drought Resilience Alliance (IDRA)	**Spain and Senegal led** 30 countries and 20 organisations.  Not a Member	• Aim: To lend **support to countries exposed to the continued threat of drought** by mobilising resources, transferring technology and knowledge, and fostering innovation. • IDRA was first announced at the UN General Assembly in September 2022 by the President of Spain.
Food and Agriculture for Sustainable Transformation Initiative (FAST)	**FAO**, in collaboration with other UN agencies.	• Aim: To implement concrete actions that would result in improving the quantity and quality of climate finance contributions to **transform agriculture and food systems by 2030,** to support adaptation and maintain a 1.5-degree pathway whilst supporting food and economic security.

Initiatives launched at COP27 of UNFCCC (Read Once)

Finance

Sustainable Debt Coalition Initiative	**Egypt**	• Aim:
		  ○ **Highlight the difficult fiscal position of emerging market and developing economies** and its debilitating impacts on climate action and development
		  ○ Spur Coalition partners to align on a set of new crucial commitments to **alleviate the debt burden.**
		  ○ **Launch a new track for consultations at the intersection of debt, climate, and development.**

Global Shield against Climate Risk	**G7 and V20** ('the Vulnerable Twenty').  • V20 is a dedicated cooperation initiative of economies systemically vulnerable to climate change. It was established in 2015 at Lima, Peru. ○ V20 Group membership stands at 58 economies.	• Aim: To provide climate **risk insurance and social protection schemes in developing countries.** • The **World Bank Group has announced a Global Shield Financing Facility** to support the initiative.
Industry Transition Programme	**Climate Investment Fund (CIF)**	• It is the **world's first large scale dedicated finance programme for developing country industry transitions.**

Initiatives launched at COP27 of UNFCCC (Read Once)

Forest and Land

Mangrove Alliance for Climate (MAC)	The **United Arab Emirates (UAE) and Indonesia.**  Member • Other members include Sri Lanka, Australia, Japan, and Spain.	• Aim: To **educate and spread awareness worldwide on the role of mangroves in curbing global warming** and its potential as a solution for climate change.
Forest and Climate Leaders' Partnership (FCLP)	**The UK and world Leaders from 26 countries** (India not a part) and **EU** launched.  Not a Member	• Aim: To **boost action to implement a commitment made by over 140 countries at COP26 in Glasgow last year to halt forest loss and land degradation by 2030** and to convert ambition into results on the ground. • Its action areas include: ○ Mobilizing public and donor finance to support implementation. ○ Supporting Indigenous Peoples' and local communities' initiatives. ○ Incentivizing conservation of high-integrity forests.

Initiatives launched at COP27 of UNFCCC (Read Once)

Forest and Land

Enhancing Nature-based Solutions for Climate Transformation (ENACT)	**COP27 Presidency, in collaboration with the IUCN and Germany**	• It will **serve as a hub for Party and non-state actors working on Nature-based Solutions (NbS)** to foster collaboration and bring global coherence to activities. • The ENACT partnership will function as an enabler and accelerator of progress towards multilaterally established global targets such as- o the **UN Decade on Restoration,** o the **proposed 30x30 target under the CBD Global Biodiversity Framework,** and o the **G20 Global Initiative on Land Degradation under the UNCCD.**

Initiatives launched at COP27 of UNFCCC (Read Once)

Health and nutrition		
Alliance for Transformative Action on Climate and Health (ATACH)	Co-convened by the **UK and Egypt**	• Aim: To realize the **ambition set at COP26 to build climate resilient and sustainable health systems** • It will use the **collective power of WHO Member States and other stakeholders** to drive this agenda forward at pace and scale; and promote the integration of climate change and health nexus into respective national, regional, and global plans.

Initiatives launched at COP27 of UNFCCC (Read Once)

Initiatives with focus on African region

Africa Just & Affordable Energy Transition Initiative (AJAETI)	**COP27 Presidency of Egypt**	• Aim:
		  ○ **Provide all Africans with access to clean energy,** whilst meeting the energy requirements for Africa's economic development.
		  ○ **Driving economic growth and supporting job creation across Africa** to build a modern, resilient, and sustainable energy systems across the continent.
		• The initiative will **engage the International Energy Agency (IEA) and International Renewable Energy Agency (IRENA).**

Decent Life for a climate resilient Africa Rationale	**President of the Arab Republic of Egypt**	Aim: To **improve the quality of life in 30% of the most vulnerable and poorest villages and rural areas** in the continent by 2030, in a climate-sensitive manner.
African Women's Climate Adaptive Priorities	**COP27 Presidency**	Aim: To **ramp up the inclusion of women for a climate-resilient future.**

Initiatives launched at COP27 of UNFCCC (Read Once)

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Initiatives with focus on African region

African Carbon Markets Initiative (ACMI)	The initiative was inaugurated at COP 27 in collaboration with the **Global Energy Alliance for People and Planet (GEAPP), Sustainable Energy for All (SEforALL), and the UN Economic Commission for Africa.**	Aim: To support the **growth of carbon credit production and create jobs in Africa.**
Friends of Greening National Investment Plans in Africa and Developing Countries initiative	**Egypt's Ministry of Planning and Economic Development** and supported by the United Nations Development Programme (UNDP)	Aim: **Shaping the process of planning and designing the economic policies in a manner that factors in the impact of climate change.**
Global Waste Initiative 50 by 2050	It will leverage **voluntary engagements from over 180 countries.**	• Aim: To **treat and recycle at least 50% of the solid waste produced in Africa by 2050.** • The initiative is **first of its kind global coalition that proposes a collaborative platform for all stakeholders involved in waste management** to holistically address all solid waste types and contribute to an ambitious target at the scale of the African continent.

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Mobility and Urban transformation

Low Carbon Transport for Urban Sustainability (LOTUS) initiative	**SLOCAT and BCG (Boston Consulting Group)** and developed in collaboration with the UN Environment Programme (UNEP)	Aim: To activate systemic change **to improve and decarbonize the urban mobility landscape.**
Accelerating to Zero (A2Z) Coalition	**UK,** the High-Level Champions, the International Council on Clean Transportation, The Climate Group, and the Drive Electric Campaign.	• Aim: To **promote and support the transition to zero emission vehicles globally.** • It's the **world's largest transportation coalition with over 200 organizations** including governments, industry and civil society. • It builds off the foundation of the **"Zero Emission Vehicles Declaration" (ZEV Declaration)** generated at COP26, which is the agreement that establishes the 2035 and 2040 zero-emission goal.
Collective for Clean Transport Finance	It is a strategic collaboration between the **High-Level Champions** and the **Smart Freight Center, World Bank, WBCSD, and the Nand & Jeet Khemka Foundation.**	• Aim: To create the **tools to change the risk profiles of investment in zero-emission transport.**

Initiatives launched at COP27 of UNFCCC (Read Once)

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Mobility and Urban transformation

Zero Emission Vehicles Emerging Market Campaign (ZEV-EM-C)	**USA**	• Aim: It is a one-year campaign that **seeks to accelerate zero-emission passenger vehicle deployment in emerging markets.**
Sustainable Urban Resilience for the next Generation (SURGe) initiative	**COP27 Presidency, in collaboration with UN-Habitat and with the facilitation of ICLEI.**	• Objective: To effectively **address some of the barriers that limit urban emissions reductions,** adapting urban systems to climate change, and building urban system resilience. • It will track **buildings and housing, urban water, urban mobility, urban waste and consumption, and urban energy**
Beat the Heat: Nature for Cool Cities Challenge	**Cool Coalition**	• Cities in developing countries are invited to participate in the challenge **by pledging to increase nature based solutions in their urban areas by 2030** and demonstrate tangible progress by 2025. • Participants will be supported via funding, technical assistance, partnership opportunities, and communications support.

Initiatives launched at COP27 of UNFCCC (Read Once)

Climate Mitigation and Adaptation Targets

First Movers Coalition (FMC) Cement & Concrete Commitment	**First Movers Coalition (FMC)** 	• First movers pledged to **purchase at least 10% near zero carbon cement and concrete by 2030.** • FMC is a coalition of companies using their purchasing power to create early markets for innovative clean technologies across eight hard to abate sectors. ○ *India is a Government Partner of the coalition.*
LeadIT (Leadership for Industry Transition) Summit 2022	**Hosted by India and Sweden** 	• The summit concluded with the adoption of the summit statement by the members of LeadIT which **re-emphasized the commitment to continue pursuing the low-carbon transition of the industry.** • LeadIT (Leadership for Industry Transition) initiative focuses on low carbon transition of the hard to abate industrial sector. ○ It was **launched by the governments of Sweden and India at the UN Climate Action Summit in September 2019** and is supported by the World Economic Forum.

Initiatives launched at COP27 of UNFCCC (Read Once)

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Climate Mitigation and Adaptation Targets

Middle East Green Initiative (MGI) Summit 2022	**Saudi Arabia led regional dialogue** to bring together leaders from across the MENA region (Middle East/North Africa) and beyond.	• **Aim:** To mitigate the impact of climate change on the region and to collaborate to meet global climate targets.
		• MGI Summit 2022 was held in Sharm El Sheikh, in tandem with COP27.
		  o MGI is a Saudi Arabia led regional dialogue on climate launched in 2021 to bring together leaders from across the MENA region (Middle East/North Africa) and beyond.
		  o *India is one of the endorsing countries.*
		• **Targets MGI:**
		  o Reducing carbon emissions from regional hydrocarbon production by more than 60%.
		  o **Plant 50 billion trees across the Middle East and restore an area equivalent to 200 million hectares** of degraded land.

Initiatives launched at COP27 of UNFCCC (Read Once)

Climate Mitigation and Adaptation Targets

Methane Alert and Response System (MARS)	**UNEP's International Methane Emissions Observatory.** • Initial financial support for MARS is provided by the European Commission, the U.S. Government, the Global Methane Hub, and the Bezos Earth Fund.	• Aim: To **accelerate implementation of the Global Methane Pledge** by transparently scaling up global efforts to detect and act on major methane emissions sources. • It is **new satellite-based system** that will alert governments, companies and operators about large methane sources to foster rapid mitigation action.
Climate TRACE (Tracking Real-time Atmospheric Carbon Emissions) Coalition	**Spearheaded by former U.S. Vice President Al Gore**	• Aim: To make meaningful climate action faster and easier by **mobilizing the global tech community to track greenhouse gas (GHG) emissions with unprecedented detail and speed** and provide this data freely to the public. • It is a **global non-profit coalition.** • It will harness **satellite imagery and other forms of remote sensing, artificial intelligence,** and collective data science

Initiatives launched at COP27 of UNFCCC

Climate Mitigation and Adaptation Targets

Sharm El-Sheikh Adaptation Agenda	• **COP27 Presidency in partnership with the High-Level Champions and the Marrakech Partnership.** ○ Underpinned by the 2,000+ organisations spanning 131 countries in the Race to Resilience campaign.	• It outlines **30 Adaptation outcomes to enhance resilience for people living in the most climate vulnerable communities by 2030.** • The outcomes include urgent global 2030 targets related to Food Security and Agriculture, Water and Nature, Ocean and Coastal, Infrastructure Systems etc.
Planning for Climate Commission	**Green Hydrogen Organisation, International Hydropower Association, the Global Wind Energy Council and the Global Solar Council.**	• Aim: It is a new global initiative focused on **speeding up planning and approvals for the massive deployment of renewables and green hydrogen** needed to address climate change and energy security.
Global Renewables Alliance	**International Geothermal Association, Global Wind Energy Council, Long Duration Energy Storage Council, Green Hydrogen Organisation, International Hydropower Association, and the Global Solar Council**	• Aim: To **position renewable energy as a pillar of sustainable development and economic growth.** • It brings together, for the first time, all the technologies required for the energy transition in order to ensure an accelerated energy transition.

Initiatives launched at COP27 of UNFCCC (Read Once)

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Climate Mitigation and Adaptation Targets

Action on Water, Adaptation and Resilience (AWARe)	**COP27 Presidency, in partnership with World Meteorological Organization (WMO)** • It is a collaboration between various stakeholders, including African Union.	• It will **address water security as part of climate change adaptation** and will focus on three priorities: o Decrease water losses worldwide and improve water supply. o Promote mutually agreed, cooperative water adaptation action. o Promote cooperation and interlinkages between water and climate action.
Insurance Adaptation Acceleration Campaign	Launched as a part of the **Race to Resilience campaign,** which is the UN-backed global campaign to catalyse a step-change in global ambition for climate resilience	• Aim: To **mobilise 3,000 insurance companies (equal to 50% of the market) by COP28 in 2023.** • The objective is to scale the industry's ability to- o **advance meaningful climate risk reduction,** and o **jointly pursue the innovative public-private partnerships** that reflect a shared mission of protecting vulnerable populations from the physical ravages of climate change.

Initiatives launched at COP27 of UNFCCC (Read Once)

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Others		
Executive Action for 2023-2027: Early Warnings for all Action Plan	**World Meteorological Organisation (WMO)**	• The plan identifies key areas for advancing universal disaster risk knowledge, and outlines the priority actions required to achieve this, building on the **Sendai Framework for Disaster Risk Reduction.** • The action plan calls for a **targeted investment of US$ 3.1 billion between 2023 and 2027** to advance four Multi-Hazard Early Warning System (MHEWS) pillars. o The funding will cover **disaster risk knowledge, observations and forecasting, preparedness and response, and communication of early warnings,** and focus on developing countries where major gaps in early warning systems remain.
Climate Responses for Sustaining Peace (CRSP)	**Cairo International Center for Conflict Resolution, Peacekeeping and Peacebuilding (CCCPA)** in its capacity as the Secretariat of the Aswan Forum for Sustainable Peace and Development.	Aim: Ensuring that **integrated climate responses contribute to sustainable peace and development** in line with national ownership and context specificity

COP28 of UNFCCC

The 2023 UN Climate Change Conference will convene from 30 November to 12 December 2023 in **Dubai, United Arab Emirates (UAE).**

It will comprise:

- the 28th meeting of the Conference of the Parties (COP 28);
- the fifth meeting of the COP serving as the Meeting of the Parties to the Paris Agreement (CMA 5);
- the 18th meeting of the COP serving as the Meeting of the Parties to the Kyoto Protocol (CMP 18)

The first Global Stocktake of the implementation of the Paris Agreement will conclude at COP 28.

India’s INDC objectives

- During Warsaw Summit 2013 (COP 19), countries agreed to publicly outline what actions they intend to take under a global agreement well before the Paris Summit 2015. These country commitments are known as Intended Nationally Determined Contributions (INDCs).

- India’s three INDC objectives were announced in October 2015 (the **Lima summit urged every country** to announce its INDCs by Nov 2015).

1. Reduce the emissions intensity of its GDP by 33-35% by 2030, below 2005 levels.
2. Achieve 40% of its total electricity capacity from renewables.
3. Create an additional carbon sink of 2.5 to 3 billion tonnes of CO_2 through afforestation by 2030

India's INDC objectives

In 2021, the Prime Minister of India had announced a new 5-point set of targets at COP-26 (Panchamrit):

1. India will increase its non-fossil fuel energy capacity to 500 gigawatt (GW) by 2030;
2. It will meet 50% of its energy requirements from renewable sources by 2030;
3. The total projected carbon emissions will be reduced by 1 billion tonnes from now through 2030;
4. The carbon intensity of its economy will be brought down to less than 45%;
5. India will achieve its target of Net Zero by 2070.

India’s INDC objectives

India’s updated NDCs

- It is prepared after carefully considering India’s national circumstances and the principle of common but differentiated responsibilities and respective capabilities (CBDR-RC).
- It represents the framework for India’s transition to cleaner energy for the period 2021- 2030.
- It updates two of the three quantitative targets of 2015 NDCs related to emissions intensity and share of non-fossil fuels in installed electricity capacity.
- Two panchamrita commitments (promises made during COP26 in Glasgow) not converted into official targets i.e,
    - 500 GW non fossil fuel energy &
    - emissions of at least one billion tonnes of carbon dioxide by 2030.

India’s INDC objectives

Targets for 2030	Previous NDC, 2015	Updated NDC, 2022
Reduce the emissions intensity of its GDP	By 33 to 35% by 2030 from 2005 level.	By 45% by 2030 from 2005 level
Cumulative electric power installed capacity About 40% from non-fossil fuel-based energy resources with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF).	40%	50%
Create an additional carbon sink through additional forest and tree cover.	2.5 to 3 billion tonnes of Co2 equivalent	Same as earlier

India's INDC objectives

Previous NDC, 2015	Updated NDC, 2022
Put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation.	Put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass movement for 'LIFE'-'Lifestyle for Environment' as a key to combating climate change.
Adopt a climate friendly and a cleaner path than the one followed hitherto by others at corresponding level of economic development.	Same

India’s INDC objectives

Previous NDC, 2015	Updated NDC, 2022
Better adapt to climate change by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management.	Same
Mobilize domestic and new & additional funds from developed countries to implement the above mitigation and adaptation actions in view of the resource required and the resource gap.	Same
Build capacities, create domestic framework and international architecture for quick diffusion of cutting edge climate technology in India and for joint collaborative R&D for such future technologies.	Same as earlier

Long-Term Low Emission Development Strategy

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- India has submitted its Long-Term Low Emission Development Strategy (LTLEDS) to the United Nations Framework Convention on Climate Change (UNFCCC) recently.
- It lays out the steps to achieve net zero carbon emission by 2070.
- Rests on 7 key transitions to low-carbon development pathways.

Low carbon pathway

- A low carbon pathway refers to a way of reducing or minimizing the amount of carbon emissions produced by human activities, particularly those associated with energy production, transportation, and industry.
- It involves transitioning to cleaner and more sustainable sources of energy, such as renewable energy sources like solar, wind, hydro, and geothermal power, as well as increasing energy efficiency and reducing energy demand.

Long-Term Low Emission Development Strategy

- Formulated under paris agreement article 4.19
- Not mandatory, unlike NDCs
- Broader in scope than NDCs as they also incorporate developmental goals.

Long-Term Low Emission Development Strategy

Elements	Current policies and targets
Low carbon development of electricity systems consistent with development	■ **NDC Target: 50% of non-fossil capacity by 2030.** ■ **Renewable Purchase Obligations** for distribution companies, open access consumers and captive power plants. ■ **Green energy corridors** to strengthen transmission networks in 8 Renewable Energy (RE) rich States. ■ **A three-fold rise in nuclear installed capacity by 2032.**
Integrated, efficient, inclusive low-carbon transport system	■ **20% ethanol blending in petrol by 2025.** ■ **Indian Railways to become net-zero by 2030.** ■ Leapfrogging to directly reach Bharat Stage VI emissions. ■ A National Master Plan for Multi-modal Connectivity – PM Gati Shakti.
Adaptation in urban design, energy and material-efficiency in buildings, and sustainable urbanisation	■ **National Urban Policy Framework (NUPF).** ■ National Building Code, Energy Conservation Building Code, Eco-Niwas Samhita (an energy conservation building code for residential buildings). ■ India Cooling Action Plan ■ **National Mission on Sustainable Habitat.** ■ Extended Producer Responsibility 2021, and Plastic Waste Management (Amendment) Rules 2021.

Long-Term Low Emission Development Strategy

Economy-wide decoupling of growth from emissions and development of an efficient, innovative low-emission industrial system	▪ **National Missions for Enhanced Energy Efficiency** and Sustainable Habitat, **Standards and Labelling Scheme,** and the **Energy Efficiency Financing Platform.** ▪ Fuel switching through promotion of natural gas and the **National Policy on Bio-Fuels.** ▪ Material efficiency through policies on resource efficiency, plastic and e-waste, and steel recycling. ▪ **Green hydrogen technology** and infrastructure promotion.
CO2 removal and related engineering solutions	▪ R&D and building human and infrastructure capacity to evolve technologies and methodologies like **Carbon Capture Utilisation and Storage** (CCUS).

Long-Term Low Emission Development Strategy

Enhancing Forest and vegetation cover consistent with socio-economic and ecological considerations	▪ **NDC target:** Create an **additional carbon sink of 2.5 to 3 billion tonnes of CO2 equivalent by 2030.** ▪ Other voluntary contributions: • **To restore 26 million ha degraded land by 2030.** • **12 National Biodiversity Targets, in line with 20 global Aichi biodiversity goals.** ▪ Major policies and institutions: National Mission for a Green India, **National Afforestation Programme, Nagar Van Yojana, National REDD+ (Reducing Emissions from Deforestation and forest Degradation) Strategy 2018** etc. ▪ Major **greening efforts of the National Highways Authority of India (NHAI) and Indian Railways.**
Economic and financial aspects of low-carbon development	▪ **Business Responsibility and Sustainability Report (BRSR) by SEBI** for mandatory sustainability reporting. ▪ Inclusion of Renewable energy projects under Priority Sector Lending (PSL). ▪ **Sustainable Finance Group established by RBI** to lead regulatory initiatives in the area of investments linked to climate risk and sustainable finance.

Measures to Regulate Ozone-Depleting Substances

Vienna Convention and Montreal Protocol

- The Vienna Convention for the Protection of the Ozone Layer (Vienna Convention) was agreed in 1985.
- It established global monitoring and reporting on ozone depletion. It also created a framework for the development of protocols for taking more binding action.
- It aimed at:
  - Promoting research and monitoring of human activities on the ozone layer.
  - Taking concrete action against activities with adverse effects on ozone layer.

Measures to Regulate Ozone-Depleting Substances

Montreal Protocol on Ozone-Depleting Substances

- The Montreal Protocol under the Vienna Convention was agreed in 1987.
- It facilitates global cooperation in reversing the rapid decline in atmospheric concentrations of stratospheric ozone (good ozone).
- Under the protocol countries agreed to phase out the production and consumption of certain chemicals that deplete ozone.
- The Vienna Convention and its Montreal Protocol are the first and only global environmental treaties to achieve universal ratification, with 197 parties.
- As a result of the international agreement, the ozone hole in Antarctica is slowly recovering because of the decrease in the overall concentration of stratospheric ODS. Climate projections indicate that the ozone layer will return to 1980 levels between 2050 and 2070.

India and Montreal Protocol

- India became Party to the Vienna Convention and the Montreal Protocol in 1991 and 1992 respectively.
- India has proactively phased out the production and consumption of CFCs except use in **Metered Dose Inhalers (MDIs)** used for treatment of Asthma and Chronic Obstructive Pulmonary Disease (COPD) ailments from 2008. Subsequently, the use of CFCs in MDIs has been phased out from 2012.
- An **Ozone Cell** has been setup (under the Environment Ministry) as a National Ozone Unit (NOU) to render necessary services for effective and timely implementation of the Montreal Protocol and its ODS phase out program in India.

Measures to Regulate Ozone-Depleting Substances

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Kigali Amendment to Montreal Protocol

- The Kigali Agreement was adopted in 2016 to phase-down hydrofluorocarbons (HFCs).
- It entered into force in 2019.
- It divides nations into 3 groups with a four-step path to achieve 80% reduction in HFCs consumption by 2047.
- It is a legally binding agreement designed to create rights and obligations in international law.
- Up till July 2021, 122 countries have ratified the Kigali amendment.
- Being under Group 2, India will develop its national strategy for phasing down of Hydrofluorocarbons by 2023 (after consultation with industry stakeholders).

Measures to Regulate Ozone-Depleting Substances

TARGETS FOR REDUCTION

	As parties (Developing countries) - Group 1	As parties (Developing countries) - Group 2	Non parties (Developing countries)
Baseline formula	Average HFC consumption levels for 2020-2022 + 65% of hydrochloroflucarbon (HCFC) baseline	Average HFC consumption levelsfor 2024-2026 + 65% of hydrochloroflucarbon (HCFC) baseline	Average HFC consumption levels for 2011-2013 + 65% of hydrochloroflucarbon (HCFC) baseline
Freeze	2024	2028	-
1st step	2029 - 10%	2032 - 10%	2019 - 10%
1nd step	2035 - 30%	2037 - 20%	2024 - 40%
3rd step	2040 - 50%	2042 - 30%	2029 - 70%
4th step			2034 - 80%
Plateau	2045 - 80%	2047 - 85%	2036 - 85%

Measures to Regulate Ozone-Depleting Substances

Notes:

1. **Group 1:** Article 5 parties not part of Group 2
2. **Group 2:** Bahrain, India, the Islamic Republic of Iran, Iraq, Kuwait, Oman, Pakistan, Qatar, Saudi Arabia and the United Arab Emirates.
3. Technology review in 2022 and every five years.
4. Technology review four to five years before 2028 to consider the compliance deferral of two years from the freeze of 2028 of Article 5 Group 2 to address growth in relevant sectors above certain threshold.

Measures to Regulate Ozone-Depleting Substances

Hydrofluorocarbons (HFCs)

- Hydrofluorocarbons (HFCs) are used as an alternative to CFCs and HCFCs/HBFCs.
- HFCs do not contribute to ozone depletion.
- However, they are potent greenhouse gases with high global warming potential.

India's position

- India has agreed on a lenient schedule as it consumes only 3% of HFCs compared to other nations like the USA (37%) and China (25%).
- India has to reduce its HFC use by 80 % by the year 2047, while China and the US have to achieve the same target by the years 2045 and 2034, respectively.
- India will complete its phase-down in 4 steps from 2032 onwards, with a cumulative reduction of 10% in 2032, 20% in 2037, 30% in 2042 and 85% in 2047.

HFO Alternatives to HFCs

- HFO stands for hydrofluoro olefin. HFO refrigerants are composed of hydrogen, fluorine & carbon atoms, but contain at least one double bond between the carbon atoms.

Fluorochemical	Ozone Depleting Potential	Global Warming Potential
(ODS + GHG) Chlorofluorocarbons (CFCs)	High	High
(ODS + GHG) Hydrochlorofluorocarbons (HCFCs)	Low	High
(GHG) Hydrofluorocarbons (HFCs)	Zero	High
(GHG) Hydrofluoro Olefin (HFOs)	Zero	Very Low

Measures to Regulate Ozone-Depleting Substances

Quito Adjustment

- In 2018, the Quito Adjustment is the adjustment made to the Montreal Protocol. It aims high to avoid 1ºC of future warming.
- In this, the decision to strengthen enforcement mechanisms of this accord in response to an unexpected rise in global emissions of the banned chemical trichlorofluoromethane or CFC-11 was adopted.
- CFC-11 is a chlorofluorocarbon (CFC), which is outlawed for almost all uses by the 1987 Montreal Protocol.
- But this gas (CFC-11) still contributes one-quarter of all chlorine reaching the stratosphere. 40 to 60 per cent of total global CFC-11 emissions are coming from China.

REDD and UN-REDD+

UN-REDD	REDD+
The UN Programme on Reducing Emissions from Deforestation and Forest Degradation (UN-REDD). **Headquarters:** Geneva, Switzerland. **Membership:** 64 Partner Countries.	Reducing emissions from deforestation & forest degradation **Role:** Conservation, sustainable management of forests and enhancement of forest carbon stocks in developing countries
It is a multilateral collaborative programme of FAO, UNDP and UNEP. It was created in 2008 in response to the UNFCCC decisions on the **Bali Action Plan and REDD.**	Voluntary climate change mitigation approach that has been developed by Parties to the UNFCCC.

REDD and UN-REDD+

UN-REDD	REDD+
The main goal of REDD is to reduce greenhouse gas emissions by incentivizing countries to preserve their forests instead of cutting them down. The program achieves this by providing financial incentives to countries that reduce their deforestation rates.	REDD+ goes **beyond deforestation and forest degradation** and aims to incentivize developing countries to reduce emissions from deforestation and forest degradation, conserve forest carbon stocks, sustainably manage forests and enhance forest carbon stocks. The "+" in REDD+ refers to these additional activities.

REDD+ also includes a **stronger emphasis on the involvement of local communities** and indigenous peoples in forest conservation and management, as well as the protection of biodiversity and ecosystem services.

Forest Carbon Partnership Facility

- It is a global partnership of governments, businesses, civil society, and Indigenous peoples focused on reducing emissions from activities commonly referred to as REDD+:

1. deforestation and forest degradation, forest carbon stock conservation,
2. the enhancement of forest carbon stocks in developing countries,
3. the sustainable management of forests.

- The World Bank assumes the functions of trustee and secretariat. The World Bank, the Inter-American Development Bank and UNDP are delivery partners under the Readiness Fund and responsible for providing REDD+ readiness support.

Forest Carbon Partnership Facility

Objectives:

1. To assist countries in their REDD+ efforts by providing them with financial and technical assistance.
2. To pilot a performance-based payment system for REDD+.
3. To test ways to sustain or enhance livelihoods of local communities and to conserve biodiversity.
4. To disseminate broadly the knowledge gained in Emission Reductions Programs (ERPs).

Climate and Clean Air Coalition (CCAC)

- In 2012, a few nations, along with the UNEP, came together to form the Climate and Clean Air Coalition.
- It is a partnership of governments, public and private sector, scientific institutions, civil society organizations, etc. committed to protecting the climate through actions to reduce short-lived climate pollutants

BioCarbon Fund Initiative

- BioCarbon Fund Initiative for Sustainable Forest Landscapes (ISFL) is a multilateral fund, supported by donor governments and managed by the World Bank.
- It seeks to promote reduced GHG emissions from the land sector, from deforestation and forest degradation in developing countries (REDD+), and from sustainable agriculture, as well as smarter land-use planning, policies, etc.

Cool Coalition

- It aims to inspire ambition and accelerate action on the transition to clean and efficient cooling.
- It was launched at the first Global Conference on Synergies between the 2030 Agenda and Paris Agreement in 2019.
- It is a global effort led by:
  1. UNEP
  2. Climate and Clean Air Coalition
  3. Kigali Cooling Efficiency Program
  4. Sustainable Energy for All (SEforALL)

Global Climate Change Alliance + (GCCA+)

- GCCA+ is a European Union initiative. It helps vulnerable countries on the front line of climate change.
- GCCA+ initiatives help mainly Small Islands Developing States (SIDS) and Least Developed Countries (LDCs) increase their resilience to climate change.
- It also supports these group of countries in implementing their commitments resulting from the 2015 Paris Agreement (COP21)

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Global Alliance for Climate-Smart Agriculture (GACSA)

- GACSA is promoted by FAO along with various governments. Its vision is to improve food security, nutrition and resilience in the face of climate change.
- GACSA works towards three aspirational outcomes to:
    a. Improve farmers' agricultural productivity and incomes in a sustainable way;
    b. Build farmers' resilience to extreme weather and changing climate;
    c. Reduce greenhouse gas emissions associated with agriculture, when possible.

GHG Protocol

- GHG Protocol is developing standards, tools and online training that helps countries, cities and companies track progress towards their climate goals.
- GHG Protocol establishes frameworks to measure and manage greenhouse gas (GHG) emissions from private and public sector operations, value chains and mitigation actions.
- GHG Protocol arose when World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD) recognized the need for an international standard for corporate GHG accounting and reporting in the late 1990s.

Arctic Council

- Arctic Council is an intergovernmental forum promoting cooperation, coordination and interaction among the Arctic states, Arctic Indigenous communities and other Arctic inhabitants on common Arctic issues, in particular on issues of sustainable development and environmental protection in the Arctic.
- The Arctic Council consists of the eight Arctic States: Canada, the Kingdom of Denmark (including Greenland and the Faroe Islands), Finland, Iceland, Norway, Russia, Sweden and the United States.

Santiago Network

- The vision of the Santiago Network is to catalyze the **technical assistance of relevant organizations**, bodies, networks and experts, for the implementation of relevant approaches for averting, minimize and addressing **Loss and Damage** at the local, national and regional level, in developing countries that are particularly vulnerable to the adverse effects of climate change.
- The Santiago Network will connect vulnerable developing countries with providers of technical assistance, knowledge, resources they need to address climate risks comprehensively in the context of averting, minimizing and addressing loss and damage.
- Mandated at **COP 25 in Madrid**, the Santiago Network will further the work of the Warsaw International Mechanism for Loss and Damage.

Warsaw International Mechanism for Loss and Damage

- The Warsaw International Mechanism for Loss and Damage Associated with Climate Change Impacts was established at the 19th session of the Conference of the Parties to the UNFCCC (COP 19) in 2013.
- It was mandated to enhance knowledge, strengthen dialogue and coordination, and enhance action to address loss and damage associated with the impacts of climate change in vulnerable developing countries.

Intergovernmental Panel on Climate Change (IPCC)

- The IPCC, the UN body for assessing the science related to climate change.
- It was established by the United Nations Environment Programme (UNEP) and the World Meteorological Organization (WMO) in 1988.
- The IPCC produces reports that support the UNFCCC.
- IPCC reports cover all relevant information to understand the risk of human-induced climate change, its potential impacts & options for adaptation and mitigation.
- The IPCC does not carry out its own original research. Thousands of scientists and other experts contribute on a voluntary basis.

Reports in News

- The work is shared among three Working Groups (WG), a Task Force and a Task Group of IPCC:
- WG I aims at assessing the physical scientific basis of the climate system and climate change.
- WG II assesses the vulnerability of socio-economic and natural systems to climate change.
- WG III focuses on climate change mitigation, assessing methods for reducing GHG emissions, and removing greenhouse gases from the atmosphere.
- The research by the WGs is published by the IPCC at regular intervals as comprehensive Assessment Reports for the understanding of human-induced climate change, potential impacts and options for mitigation and adaptation.

PYQs: Q1

2014

With reference to 'GEF', which of the following statements is/are correct?

a) It serves as financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.

b) It undertakes scientific research on environmental issues at global level

c) It is an agency under OECD to facilitate the transfer of technology and funds to underdeveloped countries with specific aim to protect their environment.

d) Both (a) and (b)

PYQs: Q1

Answer: A

- Explanation:
- GEF is an independent financial organisation (not a research body). It may fund scientific research but is not directly involved in it.
- IPCC takes care of most of the research work.

PYQs: Q2

2016

With reference to ‘Agenda 21’, sometimes seen in the news, consider the following statements:

1) It is a global action plan for sustainable development.

2) It originated in the World Summit on Sustainable Development held in Johannesburg in 2002.

Which of the statements given above is/are correct?

a) 1 only

b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

PYQs: Q2

Answer: A

Explanation:

- Agenda 21 came out as a part of Earth Summit 1992.

PYQs: Q3

Consider the below statements about Global Wildlife Programme (GWP):

1) GWP is a CITES led global partnership that promotes wildlife conservation and sustainable development

by combating illicit trafficking in wildlife.

2) The GWP is funded by the Global Environment Facility.

Which of the statements given above are correct?

a) 1 only

b) 2 only

c) Both

d) None

PYQs: Q3

Answer: b) 2 only

It is a World Bank-led & GEF-funded partnership.

PYQs: Q4

2015

What is Rio+20 Conference, often mentioned in the news?

a) It is the UN Conference on Sustainable Development

b) It is a Ministerial Meeting of the WTO

c) It is a Conference of the IPCC

d) It is a Conference of the Member Countries of the CBD

PYQs: Q4

Answer: a)

Explanation:

Earth Summit 1992 (Rio de Janeiro)- UN Conference on Environment and Development

Earth Summit 2002 (Rio+10) (Johannesburg)- World Summit on Sustainable Development

Earth Summit 2012 (Rio+20) (Rio de Janeiro)- UN Conference on Sustainable Development

PYQs: Q5

2018

The Partnership for Action on Green Economy (PAGE), a UN mechanism to assist countries in transition towards greener and more inclusive economies, emerged at (2018)

a) The Earth Summit on Sustainable Development 2002, Johannesburg

b) The United Nations Conference on Sustainable Development 2012, Rio de Janeiro

c) The United Nations Framework Convention on Climate Change 2015, Paris

d) The World Sustainable Development Summit 2016, New Delhi

Answer: b)

PYQs: Q5

Answer: b)

PYQs: Q6

2015

With reference to the IUCN and the CITES, which of the following statements

is/are correct?

1) IUCN is an organ of the United Nations and CITES is an international agreement between governments.

2) IUCN runs thousands of field projects around the world to better manage natural environments.

3) CITES is legally binding on the States that have joined it, but this convention does not take the place of national laws.

Select the correct using the code given below.

a) 1 only

b) 2 and 3 only

c) 1 and 3 only

d) 1, 2 and 3

PYQs: Q6

Answer: b) 2 and 3 only

IUCN is an NGO. CITES is an international agreement between governments (multilateral treaty).

PYQs: Q7

2011-12

Regarding "carbon credits", which one of the following statements is ***not*** correct?

a) The carbon credit system was ratified in conjunction with the Kyoto Protocol.
b) Carbon credits are awarded to countries or groups that have reduced GHGs below their emission quota.
c) The goal of the carbon credit system is to limit the increase of carbon emission quota.
d) Carbon credits are traded at a price fixed from time to time by the United Nations Environment Programme.

PYQs: Q7

Answer: D

Explanation: Carbon credit prices are traded on an exchange. So, their prices are never fixed.

PYQs: Q8

2015

Which one of the following is associated with the issue of control and phasing out of the use of ozone-depleting substances?

a) Bretton Woods Conference
b) Montreal Protocol
c) Kyoto Protocol
d) Nagoya Protocol

PYQs: Q8

Answer: B

- Bretton Woods Conference established the International Bank for Reconstruction and Development (IBRD) and the International Monetary Fund (IMF).
- **Montreal Protocol** is a legally-binding international treaty to protect the ozone layer by phasing out the production of ozone depleting substances.
- Kyoto Protocol implemented the objective of the UNFCCC to fight global warming by reducing GHG concentrations in the atmosphere to "a level that would prevent dangerous anthropogenic interference with the climate system". It is binding on the parties.
- Nagoya Protocol is an agreement to the 1992 CBD on "Access to Genetic Resources and the Fair and Equitable Sharing of Benefits".

PYQs: Q9

2015

Which of the following statements regarding 'Green Climate Fund' is/are correct?

1) It is intended to assist the developing countries in adaptation and mitigation practices to counter CC.

2) It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank.

Select the correct answer using the code given below.

a) 1 only
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

PYQs: Q9

Answer: A

- **Explanation:** GCF is founded under the aegis of UNFCCC COP

PYQs: Q10

2018

Momentum for Change: Climate Neutral Now" is an initiative launched by

a) The Intergovernmental Panel on Climate Change
b) The UNEP Secretariat
c) The UNFCCC Secretariat
d) The World Meteorological Organisation

PYQs: Q11

Answer: C

- Climate Neutral Now was launched by the UNFCCC secretariat in 2015. It is aiming at encouraging and supporting all levels of society to take climate action to achieve a climate-neutral world by mid-century, as enshrined in the Paris Agreement.

PYQs: Q12

2016

Consider the following pairs:

Terms sometimes in the news - Their origin

1) Annex-I Countries - Cartagena Protocol

2) Certified Emissions- Reductions Nagoya Protocol

3) Clean Development Mechanism - Kyoto Protocol

Which of the pairs given above is/are correctly matched?

a) 1 and 2 only

b) 2 and 3 only

c) 3 only

d) 1, 2 and 3

PYQs: Q12

2016

Answer- C) 3 only

Explanation:

Annex-I Countries, Clean Development Mechanism, Certified Emission Reductions (CERs) or carbon credits are terms associated with the Kyoto Protocol.

PYQs: Q13

2016

The term 'INDC' is sometimes seen in the news in the context of

a) pledges made by the European countries to rehabilitate refugees from the war-affected Middle East
b) plan of action outlined by the countries of the world to combat climate change
c) capital contributed by the member countries in the establishment of Asian Infrastructure Investment Bank
d) plan of action outlined by the countries of the world regarding Sustainable Development Goals

PYQs: Q13

Answer: B

- Intended nationally determined contribution (INDC) is a non-binding national plan highlighting climate change mitigation, including climate-related targets for greenhouse gas emission reductions.

PYQs: Q14

2016

With reference to the Agreement at the UNFCCC Meeting in Paris in 2015, which of the following statements is/are correct?

1) The Agreement was signed by all the member countries of the UN and it will go into effect in 2017.

2) The Agreement aims to limit the greenhouse gas emissions so that the rise in average global temperature by the end of this century does not exceed 2 °C or even 1.5 °C above pre-industrial levels.

3) Developed countries acknowledged their historical responsibility in global warming and committed to donate $ 1000 billion a year from 2020 to help developing countries to cope with climate change.

Select the correct answer using the code given below.

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

PYQs: Q14

Answer: B

- The Agreement will **not become binding** on its member states until 55 parties who produce over 55% of the world's greenhouse gas have ratified the Agreement. There is doubt whether some countries, especially the United States, will agree to do so, though the United States publicly committed, in a joint Presidential Statement with China, to joining the Agreement in 2016.

PYQs: Q14

- As of March 2019, 195 UNFCCC members have signed the agreement, and 185 have become a party to it.
- The **Paris Agreement's long-term goal** is to **keep the increase in global average temperature to well below 2 °C above pre-industrial levels**; and **to limit the increase to 1.5 °C**, since this would substantially reduce the risks and effects of climate change.

PYQs: Q15

2014

Consider the following international agreements:
1) The International Treaty on Plant Genetic Resources for Food and Agriculture
2) The United Nations Convention to Combat Desertification
3) The World Heritage Convention

Which of the above has/have a bearing on the biodiversity?
a) 1 and 2 only
b) 3 only,
c) 1 and 3 only
d) 1, 2. and 3

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2014

Answer- D) 1,2 and 3

PYQs: Q16

2015

With reference to 'Forest Carbon Partnership Facility', which of the following statements is/are correct?

1) It is global partnership of governments, businesses, civil society and indigenous peoples.

2) It provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop eco-friendly and climate adaptation technologies for sustainable forest management.

3) It assists the countries in their 'REDD+ (Reducing Emission from Deforestation and Forest Degradation+)' efforts by providing them with financial and technical assistance.

Select the correct answer using the code given below

a) 1 only

b) 2 and 3 only

c) 1 and 3 only

d) 1, 2 and 3

PYQs: Q16

Answer: C

- **Explanation:** It provides financial incentives to countries in their REDD+ efforts. There is no mention of assistance to universities, scientists.

PYQs: Q17

2015

'BioCarbon Fund Initiative for Sustainable Forest Landscapes' is managed by the

a) Asian Development Bank
b) International Monetary Fund
c) United Nations Environment Programme
d) World Bank

PYQs: Q17

Answer: D

- BioCarbon Fund Initiative for Sustainable Forest Landscapes (ISFL) is a multilateral fund, supported by donor governments and managed by the World Bank.

PYQs: Q18

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2016

What is 'Greenhouse Gas Protocol'?

a) It is an international accounting tool for government and business leaders to understand, quantify and man-age greenhouse gas emissions

b) It is an initiative of the United Nations to offer financial incentives to developing countries to reduce GHG emissions and to adopt eco-friendly technologies

c) It is an inter-governmental agreement ratified by all the member countries of the UN to reduce GHG emissions to specified levels by the year 2022

d) It is one of the multilateral REDD+ initiatives hosted by the World Bank

PYQs: Q18

Answer: A

- Greenhouse Gas Protocol is an international accounting tool for government and business leaders to understand, quantify and man-age greenhouse gas emissions

PYQs: Q19

2017

Regarding 'Global CC Alliance', which of the following statements is/are correct?

1) It is an initiative of the European Union.
2) It provides technical and financial support to targeted developing countries to integrate climate change into their development policies and budgets.
3) It is coordinated by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD).

Select the correct answer using the code given below:

a) 1 and 2 only
b) 3 only
c) 2 and 3 only
d) 1, 2 and 3

PYQs: Q19

Answer: A

- The Global Climate Change Alliance (GCCA) was launched in 2007 by the European Commission to strengthen dialogue and cooperation on climate change between the European Union (EU) and developing countries most vulnerable to climate change, in particular Least Developed Countries (LDCs) and Small Island Developing States (SIDS), which are hardest hit by the adverse effects of climate change.
- The GCCA acts as a platform for dialogue and exchange of experience between the EU and developing countries on climate policy and on practical approaches to integrate climate change into development policies and budgets.

PYQs: Q19

- The results of dialogue and exchange of views feed into the discussions on the post-2012 climate agreement under the UN Framework Convention on Climate Change (UNFCCC), and inform the technical and financial cooperation supported by the GCCA.
- Discussions take place at global, regional and national levels.
- The GCCA also provides technical and financial support to partner countries to integrate climate change into their development policies and budgets, and to implement projects that address climate change on the ground, promoting climate-resilient, low-emission development. Technical and financial cooperation, in turn, informs political dialogue and exchange of experience at regional and global levels.

PYQs: Q20

2014

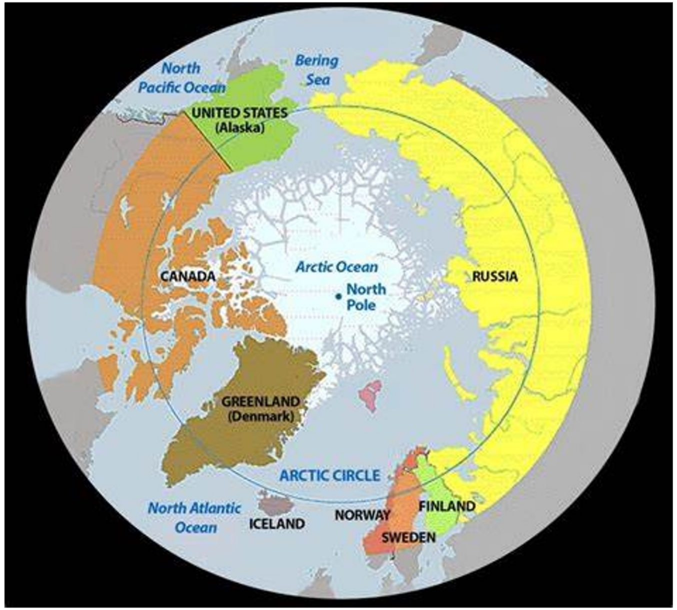
Consider the following countries:

1) Denmark
2) Japan
3) Russian Federation
4) United Kingdom
5) United States of America

Which of the above are the members of the ‘Arctic Council’?

a) 1, 2 and 3
b) 2, 3 and 4
c) 1, 4 and 5
d) 1, 3 and 5

Answer: D


North
Pacific Ocean
Bering
Sea
UNITED STATES
(Alaska)
CANADA
Arctic Ocean
North
Pole
RUSSIA
GREENLAND
(Denmark)
ARCTIC CIRCLE
North Atlantic
Ocean
ICELAND
NORWAY
FINLAND
SWEDEN